Lesson Review and Index: Product Management Lessons 1 and 2

This index covers the two reviewed lessons, the two capstone case studies, and the recommended learning order. It links each topic to the guide that teaches it in depth. These lessons did not cover RAG, Replit or Nginx in depth, so those topics have their own guides rather than being forced in here.

1. Lesson 1: AI product costing and an n8n onboarding workflow

- **Define the job** of the AI feature: tone, length, audience. Send useful variables with each request.
- **Unit cost:** calculate cost from input and output tokens, then multiply by expected volume. Compare conservative, realistic and aggressive scenarios before deciding to charge, give a free allowance, or absorb the cost. *Guide: Tokens, context and cost.*
- **Define input, output, owner and cost** before using an LLM in any feature.
- **Employee onboarding:** an HR form captures hire ID, candidate name, email, department, role, manager and start date. Reference data sits in an access registry and a scheduling registry.
- **Three agents:** a Coordinator routes and tracks work; an Access agent decides and records access; a Scheduling agent handles meetings. *Guide: n8n multi-agent onboarding.*
- **Webhooks:** a doorbell. The workflow waits for an event instead of checking repeatedly. Use the test URL while building, then the production URL once active. Include a unique `hire_id` as the primary key.
- **Credentials and tools:** never connect a personal account to a practice automation. Consider permissions before connecting any tool. Build and test one small step at a time.

2. Lesson 2: Make, MCP and case-study thinking

- **Scenarios** are repeatable automations. Variables and settings let one flow serve many inputs. Check credit use, notifications and failure handling before you enable one. *Guide: Make.com.*
- **MCP** (Model Context Protocol) is a standard way for AI applications to connect to outside systems. A **tool** is a capability an agent can call. *Guide: Skills, subagents and agents.*
- **Repeated assignment expectations:** understand the user and the real problem first; define the goal and a success metric; split features into v1, v2 and cut; define the core loop, AI

strategy and feedback; create journey maps, a PRD and three key wireframes; measure events on a dashboard.

3. Capstone: Fireflies, meeting commitments

Part	Content
Problem	Commitments made in meetings show up in summaries but are not reliably assigned, owned, tracked or closed. The class figure: only 28 percent of extracted commitments reached a resolved state within 14 days.
User	Meeting attendees who owe or receive a follow-up.
Goal	Make every commitment visible, owned and resolvable, without becoming a full project-management tool.
Resolution states	A commitment is resolved only when it is **done**, **reassigned**, or **deliberately dropped**. Each state is explicit.
Success metric	Share of commitments resolved within 14 days (the baseline is 28 percent).
v1	Detect commitments from a transcript; show owner and due date; three resolution states.
v2	Reminders; reassignment suggestions; owner summaries.
Cut	Full project boards, task dependencies, time tracking.
Core loop	Meeting → commitment extracted → owner confirms → reminder → resolved state recorded.
AI strategy	Extraction with a confidence score; the owner always confirms before a commitment is saved.
Feedback design	One-tap confirm, edit or reject on each extracted commitment, so the model learns what matters to each team.
Boundaries	No full task management; no changes to the meeting platform itself.
Events	commitment_detected, owner_confirmed, reminder_sent, resolved_done, resolved_reassigned, resolved_dropped
Wireframe 1	Meeting summary with a list of detected commitments, each with owner and due date, and confirm, edit or reject buttons.
Wireframe 2	My commitments: list grouped by due date, with the three resolution states as clear buttons.
Wireframe 3	Weekly summary: open, resolved and dropped counts, plus a link to any item that is overdue.

4. Capstone: ixigo, the confirmed-seat problem

Part	Content
Problem	A traveller knows the origin and destination but has no confirmed seat on the preferred train or class.
User segments	Last-minute travellers; price-sensitive planners; travellers who must arrive by a fixed time.
Goal	Help the user discover, compare and choose viable alternatives, and understand each option honestly.
Decision factors	Price, travel time, arrival time, station convenience, confirmation probability and cancellation flexibility.
Uncertainty messaging	Show likelihood as a range, never a guarantee. Say what is unknown and why.
v1	Compare alternative trains and classes for the same route and date; show the six decision factors; show uncertainty.
v2	Nearby stations and dates; bus and flight options where data is available.
Cut	Redesigning rail inventory, timetables, quota rules, fares, payments, refunds, or support staffing.
Success metric	Share of users who choose an option and complete the journey without leaving to search again.
Core loop	Search → see viable options → compare on the six factors → choose → track the decision.
AI strategy	Rank options and explain trade-offs in plain language; never invent availability.
Feedback design	After each choice, ask whether the option was right, and log the reason.
Events	search_started, option_viewed, option_chosen, uncertainty_shown, option_changed_later
Wireframe 1	Results list with six factor columns and a clear confidence range per option.
Wireframe 2	Option detail: trade-offs in plain language, and cancellation terms.
Wireframe 3	Confirmation: choice saved, with a reminder of what remains uncertain.

5. The nine-step case-study framework

1. Problem framing
2. User research

3. Goal
4. Success metric
5. Scope (v1, v2, cut)
6. User flow and journey map
7. PRD
8. Wireframes
9. Metrics and event taxonomy (plus an optional experiment plan)

6. Recommended learning order

1. Tokens, context and cost (LLM basics and cost).
2. RAG and retrieval.
3. Skills, subagents and agents.
4. Webhooks, JSON and test versus production.
5. One n8n workflow with fake data.
6. Credentials, permissions and test accounts.
7. Make scenarios and MCP.
8. App building with Replit or Lovable.
9. One capstone, using the nine-step framework.

7. Trusted next steps

- Build one safe n8n webhook demo with fake data and a practice account.
- Learn MCP from its introduction, then the specification when you build.
- For Make, learn scenarios first, then MCP with the smallest permissions.
- Review any connector's permissions before you connect it.

Official sources: check current documentation for n8n, Make, Lovable, Replit and MCP before relying on a feature. Prices change; check the live pricing page on the day you decide.

Tokens, Context Windows and Cost

This guide is self-contained. You do not need the class video. Every term is explained the first time it appears, every number is worked out step by step, and you can check each calculation with a calculator. It answers four questions: what a token is, how much a request costs, how to keep the context window from filling up, and how to choose a model and settings that are fast, cheap and good enough.

> **Running example.** We use one food-seller app throughout. A seller in Mumbai types a dish name, category and price, and the app writes a short, appetising description. Each button click is one API call.

1. What a token is

An LLM does not read letters or words. It reads **tokens**: chunks of text that the model's tokenizer has learned to split out. A token can be a whole short word (`dish`), part of a long word (`Separ` + `ating`), a space plus a word, or punctuation. Rule of thumb for English: **one token is roughly four characters, or about three-quarters of a word.**

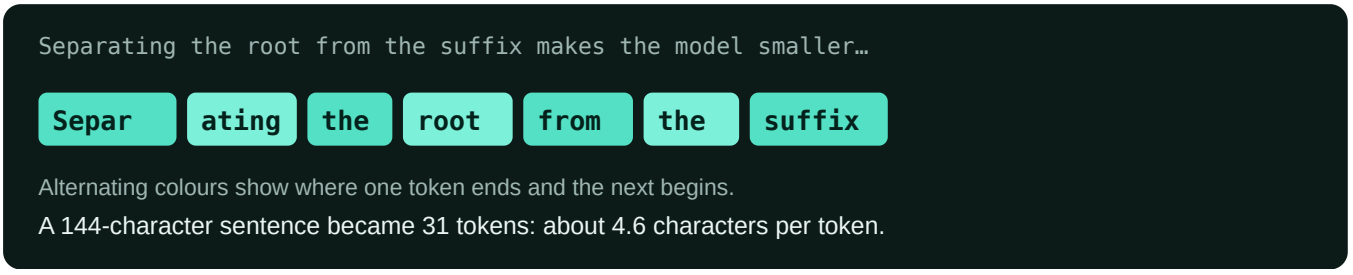


Figure 1. Tokens are not words. Common pieces are single tokens; rarer words split into several pieces. The example sentence (144 characters) is 31 tokens in one class tokenizer.

Why rare words cost more

The tokenizer was trained on a large text corpus. Frequent words such as *the* or *product* are single tokens. Rare words, names, code and non-English text break into more pieces, so they cost more tokens per idea. This is why an Indian-language or technical prompt can cost more than the same meaning in plain English. Always measure, never guess.

2. Input tokens and output tokens

Every API call has two sides, and each side is billed separately:

- **Input tokens** are everything you send: the system prompt (standing instructions), the user message, any examples, and any retrieved documents.
- **Output tokens** are everything the model writes back. Output tokens are usually priced higher than input tokens, and they are generated one by one, so they also take longer.

Total tokens per call = input tokens + output tokens. That total is what the usage report shows, and what the cost is based on.

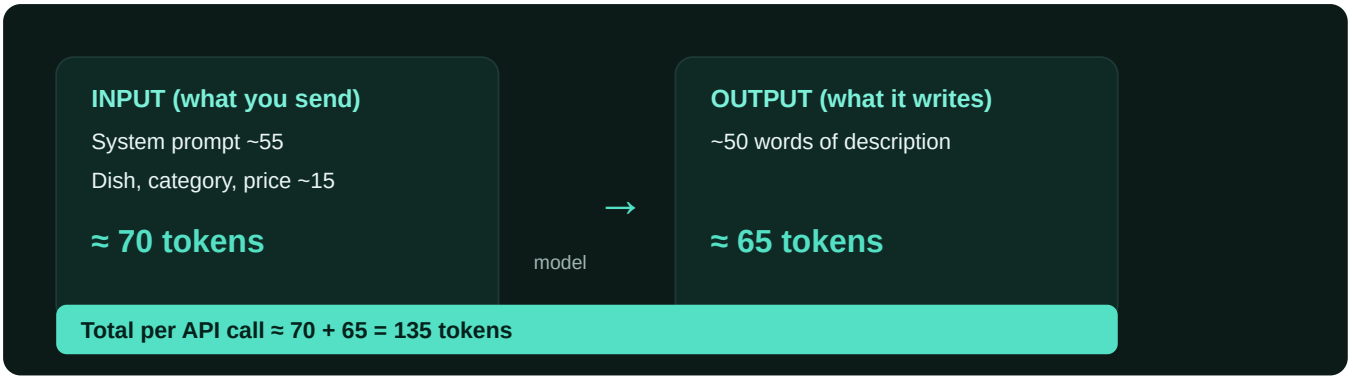


Figure 2. The seller-app call. Input is about 70 tokens, output about 65, total about 135. The system prompt is the part that repeats on every click.

3. Counting tokens before you ship

Measure first. Never estimate. Three ways to count:

1. **Count the prompt in the provider's console or token-counting endpoint.** This gives the exact input count before you pay for a call.
2. **Read the usage block in each response.** Every API response reports input and output tokens. Log them.
3. **In Claude Code or Claude Desktop, check the context usage panel** (see section 5). It shows how much of the window your session is using.

> **Common mistake:** counting only the words you typed. The system prompt, earlier turns in the conversation, examples and tool definitions are all input, and they are resent on every call.

4. Pricing, with a worked cost per call

Prices change, so the numbers below use the class example: **Claude Haiku at $0.80 per million input tokens and $4 per million output tokens.** Before you publish any cost, check the current rate on Anthropic's pricing page: anthropic.com/pricing. Use the formula, and replace the rates with today's rates.

Formula: cost per call = (input tokens ÷ 1,000,000 × input rate) + (output tokens ÷ 1,000,000 × output rate)

Worked example: one seller-app call

Step	Calculation	Result
Input cost	70 ÷ 1,000,000 × $0.80	$0.000056
Output cost	65 ÷ 1,000,000 × $4.00	$0.000260
Total per call	$0.000056 + $0.000260	**$0.000316**

So one generation costs about **$0.0003**, which is roughly 3 hundredths of a cent. The cost looks trivial, but it multiplies. Output is about 82% of the cost here, even though it is under half the tokens. That is why output length is the first thing to control.

From one call to a monthly bill

Multiply the cost per call by the number of calls. Use three scenarios so the business survives the bad case, not only the average one:

Scenario	Generations per dish	Example: 200 sellers × 15 dishes	Monthly cost (USD)
Conservative	1.2	3,000 dishes × 1.2 = 3,600 calls	3,600 × $0.000316 ≈ $1.14
Realistic	1.8	3,000 × 1.8 = 5,400 calls	5,400 × $0.000316 ≈ $1.71
Aggressive	2.8 (cap 3)	3,000 × 2.8 = 8,400 calls	8,400 × $0.000316 ≈ $2.65

The table shows the arithmetic, not a forecast. Use your own seller count and rates. Add 2 new dishes per seller per month to the dish count each month. The rule to take away: **a cost is only meaningful once it is multiplied by volume, and volume grows over time.** Decide whether to charge for generations, give a free allowance, or absorb the cost, using the conservative case as your floor.

Worked example: a multi-call feature ("Find my matches")

One click on a job-matching product can run 25 model calls. Price each call type separately, because they differ in model and size. The token counts below are **assumptions for teaching**; measure your own before you publish.

Call type	Model	Calls	Input tokens / call	Output tokens / call	Why this model
Query generation	Haiku	1	160	60	Short, structured task
Web search (external API)	Search API	4	0 tokens	0	Billed per search, not per token
Listing extraction	Sonnet	20	1,200	150	Reading messy web pages needs accuracy
Deduplication	Haiku	1	900	80	Simple comparison
Scoring and ranking	Sonnet	1	1,500	200	Judgement across the candidate's profile
Final synthesis	Sonnet	1	1,800	400	Writes the explanations users read

Cost per click, using Haiku at $0.80 / $4 and Sonnet at an *example* $3 / $15 per million (check current rates):

Call type	Input cost	Output cost	Subtotal per call × count
Query generation (Haiku, 1)	160 × $0.80/M = $0.000128	60 × $4/M = $0.000240	$0.000368
Listing extraction (Sonnet, 20)	1,200 × $3/M = $0.003600	150 × $15/M = $0.002250	$0.005850 × 20 = $0.117000
Deduplication (Haiku, 1)	900 × $0.80/M = $0.000720	80 × $4/M = $0.000320	$0.001040
Scoring (Sonnet, 1)	1,500 × $3/M = $0.004500	200 × $15/M = $0.003000	$0.007500
Synthesis (Sonnet, 1)	1,800 × $3/M = $0.005400	400 × $15/M = $0.006000	$0.011400
Total per click			**≈ $0.137 (plus search API fees)**

What this teaches: 20 extraction calls on Sonnet are about 85% of the token cost. The product lever is obvious: move extraction to a cheaper model, or trim the listing text before it reaches the model (see section 7). Notice that the search API is billed separately, so cost per click is tokens *plus* external fees.

5. The context window: your working memory

The **context window** is the total number of tokens the model can hold in one session at once: your messages, its replies, attached files, tool results and the standing instructions. Think of it as the model's desk. When the desk is full, older items must be removed or summarised.

Sizes vary by model and plan. Check the current figure in the model documentation for the model you use. Your own Claude Code usage screen in the class showed a window of about **967k tokens** for that model, so do not assume one fixed number.

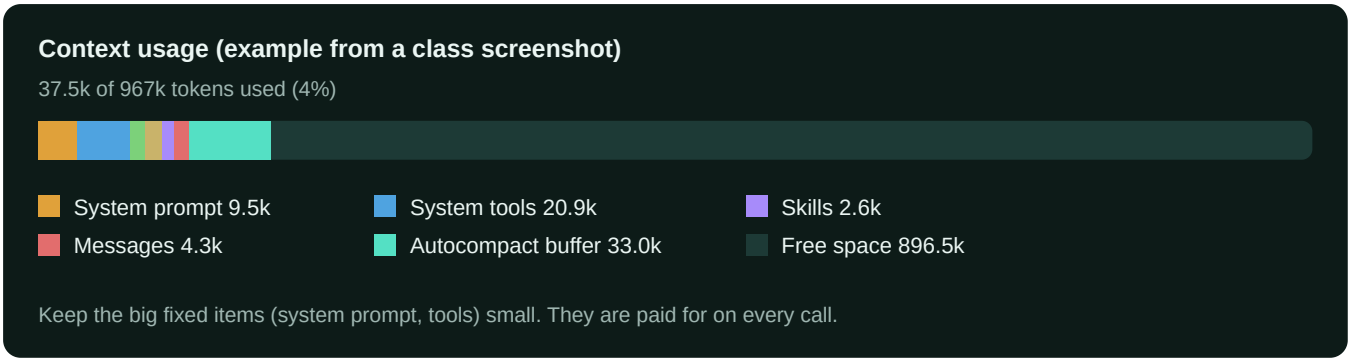


Figure 3. What fills a context window. The autocompact buffer (teal) is space kept back so the tool can summarise older messages before the window overflows.

How to check usage

- **In Claude Code:** type `/context` to see the breakdown: system prompt, tools, memory files, skills, messages and free space.
- **In an API response:** read the usage fields for input and output tokens on every call.
- **In a long chat:** start a fresh session when the topic changes. A new session starts empty, so it costs less per message.

Why replies cost more than prompts. Each reply is generated one token at a time, and it becomes part of the history that is resent with the next message. Output tokens are priced higher, so a long answer costs more than a long question. Ask for the shape you need ("five bullets, under 80 words") to keep replies short.

6. How the model chooses each word: attention and sampling

Attention: not every token gets equal weight

When the model processes the word *some* in "Riya poured tea into the cup and added some...", it does not treat every earlier word equally. It compares the current token with every earlier token, gives each a relevance score, turns the scores into weights that sum to 1, and mixes the earlier words according to those weights. In that sentence, *tea* and *cup* carry strong weight. The output

is a blend of the most relevant earlier tokens, so the same word *some* ends up with a different internal meaning depending on its context.

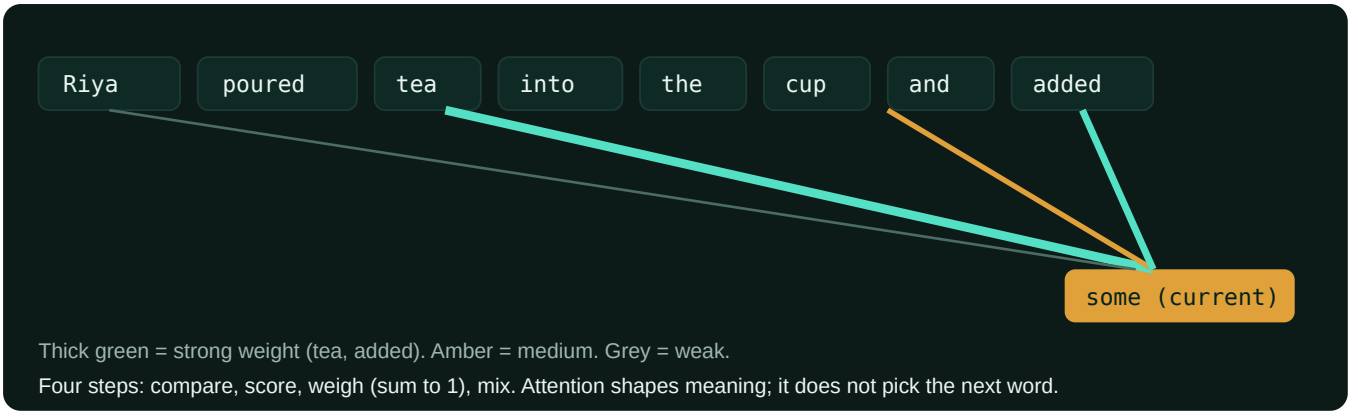


Figure 4. Attention as a weighted mix over earlier tokens. The weights are learned, not hand-written.

Sampling: how the next token is picked

After the model scores every possible next token, a sampling setting decides how the choice is made. These are the four controls you will see in most APIs:

Control	What it does	Use it when	Cost effect
Temperature	Lower = more predictable; higher = more varied. 0 is near-deterministic.	Structured output such as JSON: 0 to 0.3. Creative copy: 0.7 to 1.0.	None directly, but very high values can make output longer and less useful.
Top-k	Only the k most likely next tokens are candidates.	Stop rare, odd words from slipping in.	None directly.
Top-p (nucleus)	Keep the smallest set of tokens whose probabilities add up to p (for example 0.9).	A moderate way to balance variety and reliability.	None directly.
Max tokens	A hard limit on how many output tokens are generated.	Always set it. It caps output cost.	**Direct:** sets the largest possible output bill per call.

> **Rule of thumb for the seller app:** temperature 0.7 for the description, max tokens about 120 (enough for 50 words), and temperature 0 for any JSON the app must parse. Tune on real examples, then lock the values.

7. Using fewer tokens and getting more from each one

1. **Shorten the standing prompt, but keep the structure.** Say what the job is, the audience, the length and the format in one short block. Remove filler such as "you are a very helpful

assistant".

2. **Ask for structured output.** A JSON array of 4 search queries (about 60 output tokens) is cheaper and easier for code to use than three paragraphs of explanation.
3. **Trim large inputs before they reach the model.** Send the fields that matter, not the whole web page. Extract the job title, skills and location first, then pass only those.
4. **Use XML or clear tags to separate parts.** For example `<role>`, `<instructions>`, `<input>`. Tags help the model find each section, which reduces retries, and each retry costs tokens.
5. **Route each task to the smallest model that passes.** Use Haiku for short, structured steps and Sonnet only where judgement matters.
6. **Cap output.** Set max tokens and ask for a length ("under 50 words").
7. **Reuse, do not repeat.** Put standing instructions in a skill or project instructions, so you are not retyping them and the tool loads them efficiently (see the skills guide in this series).
8. **Cache what does not change.** Repeated prompts that share the same long system prompt can be cheaper when the provider supports prompt caching. Check your provider's current caching rules before you rely on it.
9. **Start a fresh session** when the topic changes, instead of carrying a long history into an unrelated question.

8. Choosing a model: fast, cheap or creative

Model family	Best for	Speed	Cost	Quality on hard reasoning
Haiku (small)	Classification, short copy, extraction of simple fields, query generation	Fastest	Lowest	Good for simple, well-defined steps
Sonnet (medium)	Reading messy inputs, scoring, most product features	Medium	Medium	Strong general quality
Opus (large)	Hard analysis, complex planning, high-stakes reasoning	Slowest	Highest	Strongest

Check the current list and prices in the provider documentation, since names and rates change. The decision rule: **start with the smallest model, measure quality on 20 real examples, and move up only where the smaller model fails.** Do not pay for a large model to do a simple job.

9. Limits and guardrails

- **Context limit:** the request fails or gets truncated when the window is full. Keep inputs under the limit with room for the answer.
- **Rate limit:** providers cap requests per minute. Retry with a short wait; do not retry in a tight loop, because each retry is billed.

- **Budget limit:** set a spending cap in the provider console, and log cost per feature so one bug cannot drain the budget.
- **Output limit:** max tokens stops generation early. Make sure the answer is still complete at that limit.

10. Put it into practice

> **Exercise (20 minutes).** Take any prompt you already use. (1) Count its input tokens with your provider's counter. (2) Run it once and read the output count. (3) Work out the cost with the formula in section 4 using current rates. (4) Shorten the system prompt, cap max tokens, and run it again. (5) Record the new totals and the difference in cost. Write one sentence on what you cut and whether quality held.

Quick check (answers are in the next section)

1. A reply is 65 output tokens and the prompt is 70 input tokens. Which costs more per token at the example rates?
2. Why does the system prompt matter even though you type it only once?
3. Which sampling setting caps the cost of a single call?
4. Your JSON output breaks sometimes. Which control do you lower first?

11. Answers and glossary

1. Output, at $4 per million tokens versus $0.80 for input, so output is 5 times the rate. **2.** It is resent on every call, so it is paid for on every click. **3.** Max tokens. **4.** Temperature, down towards 0.

Term	Plain meaning
Token	A small piece of text the model reads and writes, roughly four characters.
Input tokens	Everything sent to the model in one call.
Output tokens	Everything the model writes back.
Context window	The total tokens the model can hold in one session.
Attention	The model's way of weighing earlier tokens by relevance.
Temperature	How predictable or varied the chosen words are.
Max tokens	The hard limit on output length per call.
Autocompact buffer	Space kept free so older messages can be summarised before the window overflows.

12. Trusted sources to read next

- Anthropic pricing: anthropic.com/pricing (check current rates before any cost estimate).
- Anthropic documentation on models, context windows and token counting: docs.anthropic.com.
- Anthropic prompt engineering guide: docs.anthropic.com (search "prompt engineering").

Prices and model names in this guide were checked against the class material and should be re-checked on the date you publish. Token counts marked "assumption" are for teaching; measure your own before relying on them.

13. Extension: step-by-step exercises for tokens and cost

Exercise 1: count a real prompt (15 minutes)

1. Write the system prompt for the seller app on one screen, then copy it into your provider's token counter (or a text-length estimate of about 4 characters per token, to start).
2. Add one sample dish (name, category, price) and count again.
3. **You should see:** a total near the 70-token input example from section 2, or your own number.
4. Run the call once. Note the input and output counts in the usage data.

Exercise 2: cut the cost by a third

1. Shorten the system prompt by removing filler words. Keep the job, audience, length and tone.
2. Set max tokens to a value that fits your 50-word target (about 80–120 tokens).
3. Run the same three dishes before and after. Compare output tokens and read the descriptions for quality.
4. **Pass when:** output tokens fall and every description still meets the brief.

Exercise 3: choose a model

Task	Smallest model to try first	Escalate only if
Classify a dish as Breakfast, Lunch, Dinner or Dessert	Small (Haiku-class)	Accuracy is under your target on 20 examples
Write a 50-word description	Small (Haiku-class)	Tone or fluency fails the review checklist
Compare two competing descriptions and pick the better one	Medium (Sonnet-class)	Judgement is inconsistent on the test set

Current model names, limits and prices come from the provider's documentation on the day you decide. The class figures ($0.80 and $4 per million tokens for Haiku) are a historical example only.

14. Diagram: where the tokens go

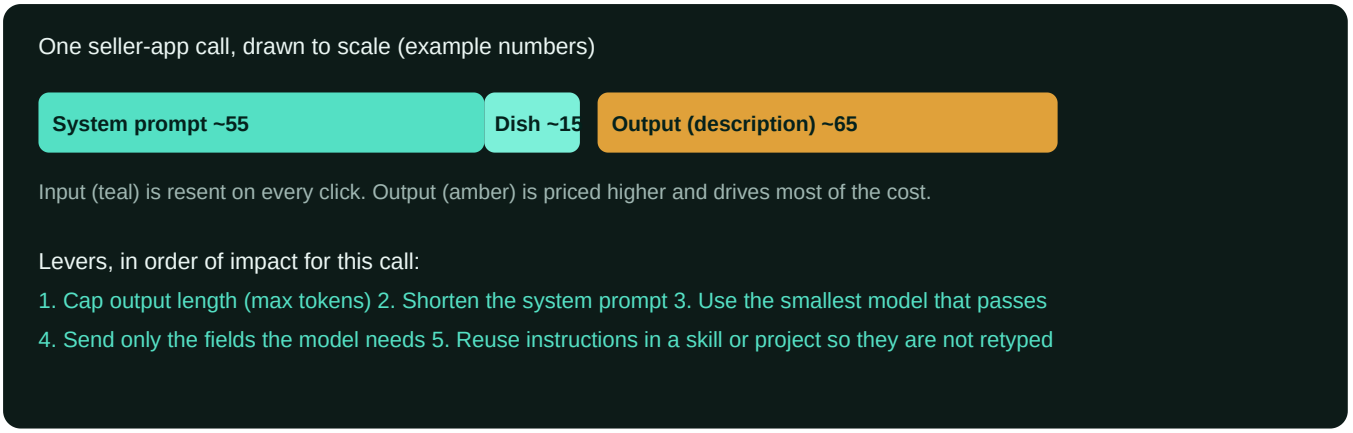


Figure 4. Token budget per call, with the levers ranked by impact.

15. Advanced: prompt caching and batching (concepts)

- **Caching:** if many requests share the same long instructions, some providers can reuse that part at a lower rate. Check the provider's current rules before you design around it.
- **Batching:** sending many non-urgent requests together, where the provider offers it, can lower cost. It adds delay, so use it only where waiting is acceptable.
- **Structured output:** asking for JSON with a fixed schema shortens output and makes the next step reliable.

16. Extended answer key and checklist

Item	Done
Counted input and output for one real call	[ ]
Set max tokens on every call	[ ]
Chose the smallest model that passes a test set of 20	[ ]
Worked the cost per call with current rates	[ ]
Logged usage for every feature	[ ]

17. Full walkthrough: count your tokens, step by step

1. **Open your provider's token-counting tool** (a console page or an official counting endpoint). Use the official tool, not a guess.
2. **Paste only the system prompt.** Note the count. This is the part that repeats on every call.
3. **Add one realistic user message** (for example "Dish: Masala Dosa, Category: Breakfast, Price: Rs 80"). Note the new total.
4. **Subtract** to find the user message's own share: total minus system prompt.
5. **Run the call once** and read the usage fields for input and output tokens. Write both down.
6. **Compare** your counted input with the usage field. Small differences are normal (formatting, hidden instructions), large ones mean you missed a part of the request.

> **You should see:** a system prompt count, a user-message count, and an output count that roughly matches the description length. Keep the three numbers in a log; this log is your evidence for every cost decision.

18. Full walkthrough: read the context usage panel

Claude Code shows a context usage panel (run `/context` to open it). The class example showed about 37.5k tokens used out of 967k, which is 4 percent. Read it in this order:

Line	What it means	What you can change
System prompt	The tool's own standing instructions	Nothing directly; it is fixed by the tool
System tools	Descriptions of the tools available to the session	Disable tools you do not use
Custom agents	Definitions of your subagents	Keep definitions short
Memory files	Standing notes such as CLAUDE.md	Keep only facts the project needs
Skills	Skill descriptions loaded for this session	Remove skills you are not using
Messages	The conversation so far	Start a new session when the topic changes
Autocompact buffer	Space kept back so older messages can be summarised	Not directly; it protects the session
Free space	What remains for new work	Monitor it on long tasks

Rule: the fixed parts (system prompt, tools, memory, skills) are paid for on every turn. Trim them before you trim the conversation.

19. Attention, explained with a sentence

Take the sentence "Riya poured tea into the cup and added some". When the model reaches the word *some*, it does four things, in order:

1. **Compare:** the current token is compared with every earlier token.
2. **Score:** each earlier token gets a relevance score for this comparison.
3. **Weigh:** the scores are turned into weights that add up to 1. *tea* and *cup* get large weights; *the* gets a small one.
4. **Mix:** the earlier words are blended using those weights to form the new internal representation of *some*.

Contextual vector: a word's starting representation is the same wherever it appears, but after attention it is shaped by its neighbours. So *some* in this sentence is no longer a generic "some"; it carries the context of tea and cup. This is why the same word can mean different things in different sentences.

> **Attention does not choose the next word.** It improves the model's understanding of the current context. The next word is chosen later, by sampling (section 6).

20. Sampling, with a worked example

After the model scores every possible next token, sampling decides which one is written. Here is one small example with three candidate words for the description "A warm bowl of ___":

Candidate	Score (before temperature)	Probability at temperature 1.0	Probability at temperature 0.3
comfort	2.0	0.66	0.96
flavour	1.0	0.24	0.03
potato	0.0	0.09	0.00

Lower temperature makes the top candidate much more likely, so output is more predictable. Higher temperature spreads the probability, so output is more varied. For JSON the app must parse, use a low value. For a short creative description, a moderate value (around 0.7) usually works. Numbers here are illustrative; tune on your own examples.

- **Top-k** keeps only the k most likely candidates. With k = 3 in this example, only the three words above are eligible.
- **Top-p** keeps the smallest set of candidates whose probabilities add to p. With p = 0.9, "comfort" and "flavour" (0.90 together) would be enough.

- **Max tokens** stops generation at a fixed length, whatever the sampling settings.

21. Model selection framework

Score each candidate model on six criteria before choosing. A simple 1 to 3 scale is enough (1 = weak, 3 = strong).

Criterion	Question to ask	How to measure
Quality	Does it pass your test set?	Score 20 real examples against the checklist
Speed (latency)	How long does one answer take?	Median time over 10 runs
Cost	What does one answer cost?	Tokens times current price (section 4 formula)
Privacy	Where does the data go, and under what terms?	Read the provider's data policy for your plan
Task risk	What happens if it is wrong?	Low risk: small model; high risk: stronger model plus human review
Context need	Does the task need a long input?	Check the model's current context limit in its documentation

Decision rule: start with the smallest model, measure on 20 examples, and move up only where it fails. Record the scores so the choice can be explained later. Current model names, limits and prices come from the provider's documentation on the day you decide.

22. Full cost worked example, with the class growth assumptions

The class scenario assumed 200 sellers at launch, 500 at month 3, 1,500 at month 6 and 5,000 at month 12, with 15 dishes per seller at onboarding and 2 new dishes per seller each month. Use the same structure with your own numbers:

1. **Dishes per month** = sellers × dishes per seller (onboarding dishes once, then 2 new a month).
2. **Calls per month** = dishes × generations per dish (conservative 1.2, realistic 1.8, aggressive 2.8, capped at the product limit of 3).
3. **Cost per month** = calls × cost per call (section 4 formula, using current rates).

Month	Sellers	Dishes added (approx.)	Calls at realistic (1.8)	Cost at $0.000316 per call (class example)

1	200	3,000 onboarding	5,400	about $1.71
6	1,500	about 3,000 new dishes a month	about 5,400 a month, plus onboarding	scale with the same formula
12	5,000	about 10,000 a month	about 18,000 a month	scale with the same formula

The per-call cost is a class example, not a current price. Replace it with the current rate before you use any number. The purpose of the table is the method: volume multiplies cost, so a small per-call price can still need a budget plan.

23. Find my matches: every call, with editable inputs

A single click can run many calls. List every call, its model, its input and output sizes, and its unit cost. Leave any vendor price as an editable input until you have the current official figure.

Call	Model	Input tokens	Output tokens	Count	Unit cost
Query generation	Small	160	60	1	input rate × 160 + output rate × 60
Web search	External search API	n/a	n/a	4	[editable: current search price]
Listing extraction	Medium	1,200	150	20	input rate × 1,200 + output rate × 150
Deduplication	Small	900	80	1	input rate × 900 + output rate × 80
Scoring and ranking	Medium	1,500	200	1	input rate × 1,500 + output rate × 200
Final synthesis	Medium	1,800	400	1	input rate × 1,800 + output rate × 400

Lesson: the 20 extraction calls usually dominate. Cutting their input (send only the title, skills and location) is the biggest lever. The token counts are teaching assumptions; measure your own.

24. Token-saving techniques with a before and after

Before (about 60 tokens of system prompt)

```
You are a very helpful and friendly assistant who writes wonderful descriptions for
food sellers. Please make sure the description is appetising and nice, and please try
```

```
to keep it short if you can, maybe around fifty words or so. Thank you very much.
```

After (about 25 tokens, same job)

```
Role: menu copywriter for local food sellers.
Task: 50-word appetising description.
Tone: warm. No prices, no health claims.
Output: plain text only.
```

The rewritten version keeps every rule and removes the politeness and repetition. Test both on the same five dishes; if quality holds, keep the shorter one.

Other techniques

- **Structured output:** ask for JSON with named fields instead of prose the code must parse.
- **Trim inputs:** pass the fields that matter, not whole web pages.
- **Retrieval:** send only the relevant passages (see the RAG guide) instead of a whole document.
- **Reusable skills and project instructions:** store standing rules once instead of retyping them.
- **XML or clear tags:** separate role, instructions and input so the model does not need to guess which part is which, which reduces retries.
- **Fresh sessions:** start a new session for a new topic.

25. Glossary (complete)

Term	Plain meaning
Token	A small piece of text, roughly four characters in English.
Tokenizer	The part that splits text into tokens.
Input tokens	Everything sent to the model in one call.
Output tokens	Everything the model writes back.
Context window	The total tokens the model can hold at once.
Autocompact buffer	Space kept back so older messages can be summarised.
Attention	Weighing earlier tokens by relevance.
Contextual vector	A token's representation after attention, shaped by its context.
Temperature	How predictable or varied the chosen word is.
Top-k, top-p	Limits on which candidate words are eligible.

Max tokens	A hard cap on output length.
Prompt caching	Reuse of a repeated prompt part, where the provider supports it.

26. Practice quiz (answers follow)

1. A prompt is 80 input tokens and the reply is 120 output tokens. Which costs more at a rate of $0.80 input and $4 output per million? Work it out.
2. Which setting caps the cost of one call most directly?
3. Why does lowering temperature help a JSON step?
4. What does the autocompact buffer protect?
5. Why is the system prompt worth trimming even though you type it once?

Answers: 1. Input: 80 × $0.80 / 1,000,000 = $0.000064. Output: 120 × $4 / 1,000,000 = $0.00048. Output costs more. 2. Max tokens. 3. Lower temperature makes the highest-scoring token much more likely, so the format is more reliable. 4. Space for summarising older messages before the window overflows. 5. It is resent on every call, so it is paid on every click.

27. Full checklist

Item	Done
Counted system prompt and one user message with the official counter	[ ]
Read the usage fields for input and output on a real call	[ ]
Set max tokens on every call	[ ]
Chose temperature by task (low for structured, moderate for creative)	[ ]
Scored candidate models on the six criteria	[ ]
Worked cost per call and per month with current rates	[ ]
Listed every call in a multi-call feature, with editable vendor prices	[ ]
Trimmed fixed parts of the context (tools, memory, skills)	[ ]

SkillNavora GenAI Launchpad · Written by Varun Kumar · Part 2 of the LLM and agents series

RAG: Giving an LLM Your Own Knowledge

This guide takes you from the problem to a working retrieval pipeline. Read it in order. Every concept is explained the first time it appears, and each section ends with a plain-English summary you can reuse. You do not need the class video.

> **Running example.** A company has a 40-page leave policy. An employee asks: *"How does our leave policy work?"* We follow that one question from the document, through the database, to the answer.

1. Three problems with LLMs

A large language model learned from a big public snapshot of text, and then stopped learning. That causes three problems:

Problem	What it looks like	Example
1. It cannot see your private data	It has never read your files.	"How many leave days do I get?" It guesses a typical answer.
2. Knowledge cutoff	Its knowledge stops at a training date.	"What changed in last week's policy?" It cannot know.
3. Hallucination	It answers confidently, and the answer is wrong.	It invents a clause that does not exist.

Three ways to fix these are covered below: fine-tuning, in-context learning and RAG. They are not interchangeable, and choosing the wrong one is the most common mistake.

2. Fine-tuning: retraining the model

Fine-tuning means continuing the model's training on your own examples, so its internal weights change. Think of it as sending the student back to school.

Three types

- **Supervised fine-tuning (SFT):** you give input and ideal output pairs, and the model learns to copy that style and behaviour.

- **Continued pretraining:** the model keeps reading large amounts of new raw text in your domain, for example medical papers.
- **RLHF (reinforcement learning from human feedback):** people rank answers, and the model learns to prefer the ones people rank higher. This is how many chat models are tuned for helpfulness.

How supervised fine-tuning works, step by step

1. **Data collection:** gather hundreds to thousands of clean examples. Quality matters more than quantity.
2. **Method selection:** choose SFT, a parameter-efficient method, or RLHF, based on your goal.
3. **Training:** run the job on a training platform and watch the loss curve.
4. **Evaluation:** test on examples the model never saw, and compare with the base model.

What fine-tuning fixes

Problem	Solved?	Why
Private data	Solved, partly	The model learns patterns from your data, but it can still mix up facts and cannot cite a source.
Recent data	Partly	Every new fact needs another training run.
Hallucination	Partly	Tuning on good answers can reduce invention, but it does not guarantee correct facts.

3. In-context learning: teaching in the prompt

In-context learning means the model learns a pattern from examples inside the prompt, with **no change to its weights**. Nothing is retrained. The learning lasts only for that request.

Few-shot example: sentiment

```
Below are examples of text labelled with their sentiment.
Use the examples to determine the sentiment of the final text.

Text: "I love this phone. It's so smooth." → Sentiment: Positive
Text: "This app crashes a lot."            → Sentiment: Negative
Text: "The camera is amazing."             → Sentiment: Positive
Text: "I hate the battery life."           → Sentiment: ???
```

The model answers **Negative**. It learned the label pattern from three examples, with no training. This is called **few-shot prompting**. Zero examples is zero-shot, and one example is one-shot.

> **Limit:** every example costs input tokens on every call (see the tokens guide). In-context learning is cheap to try and expensive to scale when the examples are long.

4. RAG: retrieval-augmented generation

RAG means: before the model answers, a search step finds the most relevant passages from your documents, and those passages are placed into the prompt. The model then answers using both its general knowledge and the passages you supplied.

Use the open-book exam picture. An LLM with fine-tuning is a student who studied the textbook months ago. An LLM with RAG is a student who is allowed to open the textbook to the right page during the exam.

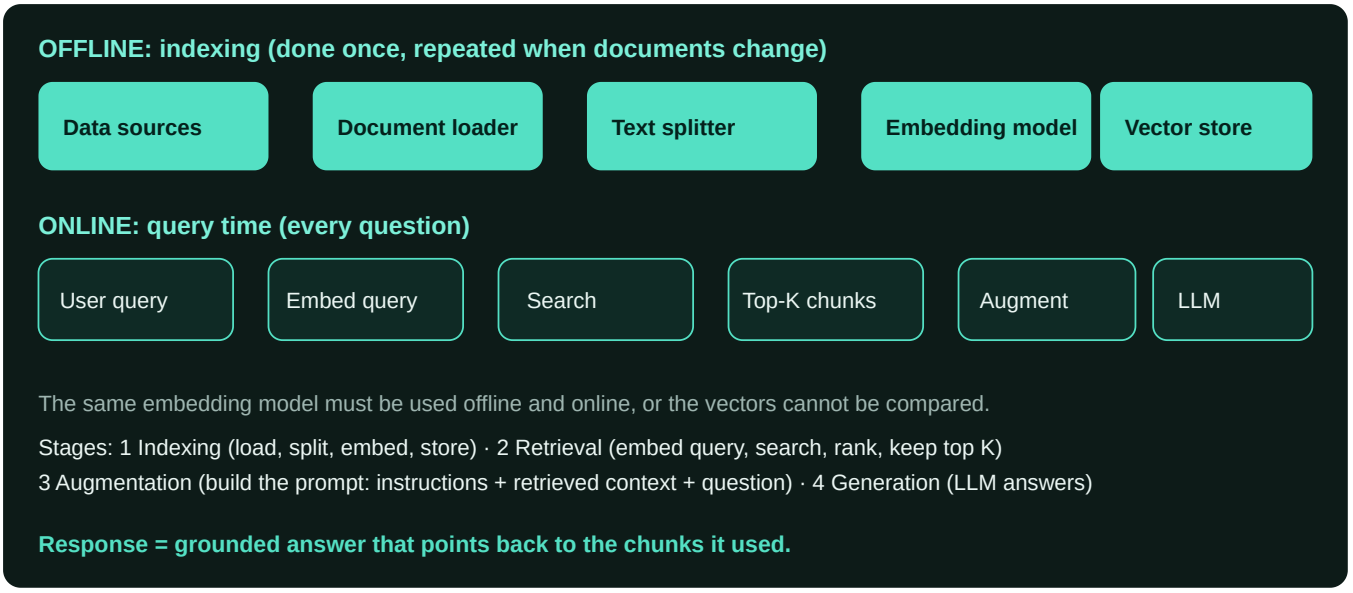


Figure 1. The complete RAG pipeline: an offline indexing half and an online query half.

5. Stage 1: indexing (the offline half)

Step 1: data sources and document loaders

A loader reads each source and turns it into plain text with metadata (file name, page, date, URL). Common sources: PDFs, web pages, YouTube transcripts, cloud storage folders, databases, CSV files. Keep the metadata: you will need it later to filter results and to cite sources.

Step 2: text chunking

A model cannot search a 40-page file as one unit. The text is cut into **chunks**. Each chunk needs to be small enough to be specific, and large enough to hold a complete thought.

Worked example: recursive character splitting

A recursive splitter tries the largest separator first, and only falls back to a smaller one when a piece is still too long:

1. Paragraph break (`\n\n`)
2. Sentence end (`. `)
3. Comma (`, `)
4. Word (space), and as a last resort single characters

Two settings control the result. **Chunk size** is the maximum length of each chunk. **Chunk overlap** is how many characters the next chunk repeats from the end of the previous one, so a sentence cut in half still appears whole in one of the chunks.

The class demo used a 519-character paragraph about machine learning, with chunk size 120 and overlap 15. The first chunk came out as *"Machine learning is a subset of artificial intelligence that focuses on building systems that learn from data"* (about 109 characters). The next chunk started with *"learn from data. Unlike traditional programming..."*, so the words "learn from data" appear in both chunks. Seven chunks came out in total.

Setting	Too small	Too large
Chunk size	Each chunk loses context; answers are fragments.	Search becomes vague; too much irrelevant text goes to the model.
Overlap	Ideas cut at a boundary get lost.	Many duplicate chunks; more storage and cost.

> **Starting point:** chunk size 300 to 800 tokens with 10 to 15 percent overlap for prose policies. Test on 20 real questions and adjust.

Step 3: embeddings: turning meaning into numbers

An **embedding** is a list of numbers (a **vector**, often hundreds or thousands of values) that represents the meaning of a piece of text. Texts with similar meaning get vectors that point in similar directions. An **embedding model** produces these vectors. Common choices include OpenAI's embedding models and open sentence-transformer models, and hosted options such as Google's and Hugging Face's models.

Rule: the same embedding model must be used for the chunks and for the questions. Vectors from different models are not comparable.

Step 4: the vector store

A **vector database** stores each chunk with three things: the chunk text, its embedding vector, and its metadata. It can run locally (for a laptop demo or a small internal tool) or in the cloud (for

shared, large or production systems). Examples include local libraries and managed cloud services; choose one by data size, privacy needs and budget.

6. Stage 2: retrieval (the online half)

Retrieval finds the right context for one question.

1. **Embed the query** with the same embedding model.
2. **Semantic search:** compare the query vector with every chunk vector, usually with **cosine similarity** (how closely two vectors point the same way; 1.0 is identical direction).
3. **Rank** the chunks by similarity.
4. **Keep the top K** chunks. K is the number of chunks you pass on, often 3 to 8.

Retrieval in action

Question: *"How does our leave policy work?"* The vector store holds chunks such as "Office dress code policy", "Annual leave and holiday rules", "Remote work equipment", "Leave policy: sick and casual", and "Employee onboarding steps". Their similarity scores to the question might be 0.91 for annual leave, 0.88 for sick and casual leave, and much lower for the rest. With K = 2, the two leave chunks are passed on. The search matched **meaning**, not the exact words "leave policy", which is why semantic search beats keyword search for questions phrased in different ways.

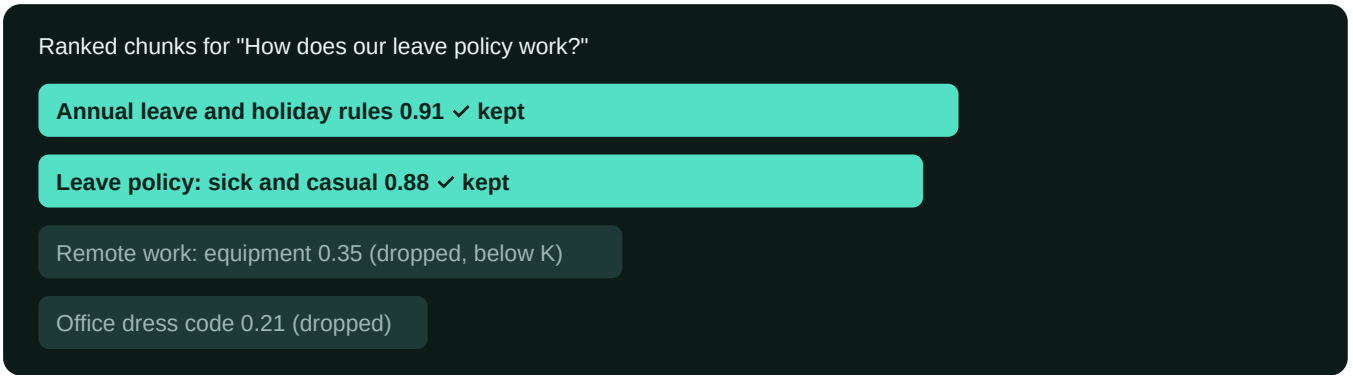


Figure 2. Retrieval keeps the top K. Scores are illustrative.

7. Stage 3: augmentation (building the prompt)

Augmentation is where you combine the instructions, the retrieved text and the question. This is the step that makes RAG work:

```
[System instruction]
You are a helpful assistant. Answer ONLY from the provided context.
If the context is insufficient, say "I don't know."
```

```
[Context]
{retrieved_chunks_text}

[Question]
{user_query}
```

Three rules make this safe. Put the rule about "answer only from context" in the system instruction. Label the context clearly so the model knows what is evidence. Ask the model to name the chunk it used. Without the instruction, the model will fall back to its general knowledge and you will not know which answers came from your documents.

8. Stage 4: generation

The LLM writes the answer from the augmented prompt. Good answers follow three checks: they are supported by the context, they say "I don't know" when the context is missing, and they cite the source. Log the question, the retrieved chunks and the answer, so you can review failures later.

9. How RAG solves the three problems

Problem	How RAG helps	What it does not fix
Private data	Your documents are searched at query time; the model sees them.	Access control: you must still decide who may see which document.
Knowledge cutoff	Re-index a document and the answer updates without retraining.	Re-indexing is still a pipeline you must run and monitor.
Hallucination	The answer is grounded in retrieved text, and you can check it.	Bad retrieval still produces bad answers, so you must measure retrieval.

10. RAG or fine-tuning? Questions and answers

Q11. A formal Hindi chatbot: RAG or fine-tuning?
Start with prompting for tone and language, which is cheap. Use RAG for facts such as policies and prices. Fine-tune only if the style still fails after good prompting and you have enough high-quality examples. The usual answer is RAG plus a clear system prompt, not fine-tuning.

Q12. Can you use RAG and fine-tuning together?
Yes. They do different jobs. Fine-tuning can teach a consistent style or format. RAG supplies

the facts for each question. Many production systems do exactly this, but add fine-tuning last, after RAG is measured.

Q13. RAG for real-time news from the last hour: suitable?
RAG can work only if the index is refreshed quickly enough, with a pipeline that loads and embeds new articles within minutes. Fine-tuning is the wrong tool, because it cannot keep up. Measure the delay between publication and availability in the index.

Q14. RAG over 500 PDFs gives partly correct answers. What is wrong?
Usually retrieval, not the model. Check: are the right chunks in the top K? If not, the chunks may be too large or too small, K may be too low, the PDFs may be scanned images with bad text extraction, or the metadata filter may be too narrow. Fix retrieval first; the model cannot answer from chunks it never received.

Q15. RAG works in English but fails in Hindi. Why?
The embedding model is probably not strong in Hindi, so Hindi questions and Hindi chunks land far apart in vector space. Use a multilingual embedding model, index the Hindi text in the same space, and test with Hindi questions. Check tokenization too: Hindi text often uses more tokens for the same meaning, which affects chunk size.

11. A test plan you can follow

1. Write 20 real questions and the answer you expect, with the source page.
2. For each question, log the top K chunks. Is the right source in the top K? (This is **retrieval accuracy**.)
3. Check whether the final answer is supported by the chunks. (This is **faithfulness**.)
4. Ask 3 questions the documents cannot answer. The correct response is "I don't know".
5. Change one setting at a time (chunk size, overlap, K, embedding model) and re-run the 20 questions.

12. Quick check

1. Which of the four RAG stages happens offline?
2. Why must the same embedding model be used for chunks and queries?
3. Name two things you should store with each chunk.
4. An answer is wrong, but the right chunk was in the top K. Is retrieval the problem?

Answers: 1. Indexing (load, split, embed, store). 2. Vectors from different models cannot be compared. 3. Its text and its metadata, and its vector. 4. No; the chunk was retrieved, so the problem is the prompt or the model's use of it.

13. Glossary and further reading

Term	Plain meaning
RAG	Search your documents first, then answer with the results in the prompt.
Chunk	A piece of a document small enough to search and send.
Embedding	A list of numbers that represents meaning.
Vector store	A database that stores chunks, their vectors and metadata.
Cosine similarity	How closely two vectors point the same way.
Top K	How many best-matching chunks are passed to the model.
Few-shot	Giving examples in the prompt so the model copies the pattern.
Fine-tuning	Retraining the model's weights on your examples.
Hallucination	A confident answer that is not supported by facts.

- Anthropic documentation on prompting and long-context use: docs.anthropic.com.
- Embedding model and vector database documentation from your chosen provider; check current options before you pick one.

13. Extension: evaluating a RAG system

Measure each stage separately. Most RAG failures come from retrieval, and you cannot fix what you do not measure.

Measure	Question it answers	How to measure simply
Retrieval recall	Is the right chunk in the top K?	For 20 questions, mark whether the source is in the top K (yes or no). Report the share of yes.
Grounding (faithfulness)	Is every claim supported by the retrieved text?	Read each answer; mark any sentence not found in the chunks.
Abstention	Does it say "I don't know" when the context is missing?	Ask 3 unanswerable questions; count correct abstentions.
Latency	How long does one answer take?	Time from question to answer, median of 10 runs.
Cost	What does one answer cost?	Input tokens (question plus K chunks) and output tokens, priced with the tokens-guide formula.

Failure modes and fixes

Symptom	Likely cause	First fix to try
Right document, wrong passage	Chunks too large or too small	Change chunk size; keep 10–15 percent overlap
Right passage not retrieved at all	K too small or wrong embedding model	Raise K; test another embedding model
Answer adds facts not in the context	Instruction too weak	Tighten the augmentation prompt; ask for citations
Hindi or regional-language questions fail	Embedding model not multilingual	Use a multilingual embedding model; test with native-language questions
Recent documents missing	Index not refreshed	Add an ingestion schedule; log the last index time
Slow answers	Too many chunks or a large model	Lower K; use the smallest model that passes

14. Diagram: the augmented prompt, labelled

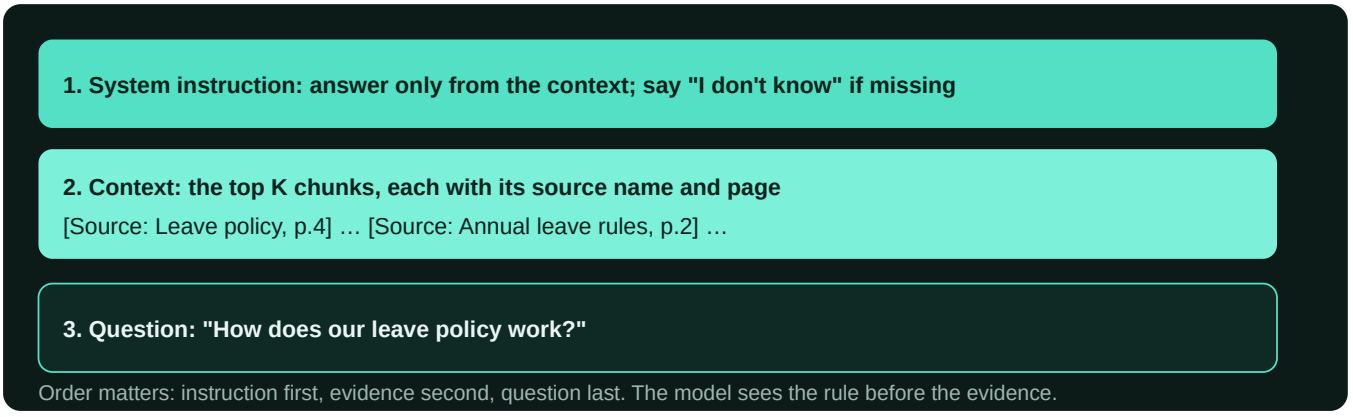


Figure 5. The three blocks of an augmented prompt.

15. Advanced: a retrieval test set you can reuse

```
question_id, question, expected_source, expected_answer_points, notes
Q01, "How many annual leave days do I get?", Annual leave rules p.2, "number of days; carry-over rule", normal
Q02, "Can I take sick leave without a note?", Sick and casual p.1, "notice rule; note rule", normal
Q03, "What is the office dress code?", Office dress code p.1, "should be retrieved, not leave", distractor check
Q04, "What is the CEO's salary?", none, "must say I don't know", abstention check
```

```
Q05, "Leave policy in Hindi", Leave policy p.1, "same answer as English",
multilingual check
```

Run the set after every change (chunk size, overlap, K, embedding model, prompt). Keep a log so you can see which change helped and which hurt.

16. Extended answer key and checklist

Item	Done
Test set of at least 20 questions with expected sources	[ ]
Retrieval recall measured before tuning the prompt	[ ]
Abstention tested with unanswerable questions	[ ]
Multilingual questions tested if your users need them	[ ]
Index refresh schedule set and logged	[ ]

17. The three problems, in depth

Problem	Why it happens	What RAG does	What RAG cannot do
Private data	The model was never trained on your files	Searches your files at question time and passes the relevant text in	Decide who may see which file; that is access control
Knowledge cutoff	Training stops at a date	Answers from the latest index you built	Know facts that were never indexed
Hallucination	The model predicts plausible text when it lacks facts	Gives it the facts, and lets you check the source	Stop a bad retrieval from producing a wrong answer

Limits matter. RAG reduces hallucination; it does not remove it. Measure grounding and abstention (section 13 of the extended guide) before you trust an answer.

18. Fine-tuning versus in-context learning versus RAG

Approach	Changes the model?	Needs retraining?	Best for	Weak at
In-context learning	No	No	Teaching a pattern with a few examples	Large amounts of knowledge; cost per call

				grows with examples
RAG	No	No (re-index instead)	Answers from changing documents with sources	Style and behaviour changes
Fine-tuning	Yes (weights change)	Yes	Consistent style, format, or a narrow task	Fresh facts; every new fact needs another run

Sentiment example (few-shot, in-context learning)

```
Text: "I love this phone. It's so smooth." → Sentiment: Positive
Text: "This app crashes a lot."            → Sentiment: Negative
Text: "The camera is amazing."             → Sentiment: Positive
Text: "I hate the battery life."           → Sentiment: ???
```

The model answers Negative. It learned the labelling pattern from three examples, with no training. Each example costs input tokens on every call, so keep them few and short.

19. Fine-tuning steps, in plain language

1. **Data collection:** gather examples of the input and the ideal output. Remove personal data. Quality beats quantity.
2. **Method selection:** choose supervised fine-tuning for format and style, continued pretraining for new domain text, or RLHF for ranked preferences.
3. **Training:** run the job on a training platform and watch the loss curve for signs of overfitting.
4. **Evaluation:** test on examples the model never saw, and compare with the base model using the same checklist.

Rule: try prompting first, then RAG, then fine-tuning. Each step is more expensive to build and maintain.

20. Chunking, worked through

A recursive splitter tries separators in order: paragraph, sentence, comma, then word. Two settings control the output.

Setting	Example value (teaching)	Effect
Chunk size	120 characters	Each chunk is at most about 120 characters

Chunk overlap	15 characters	The next chunk repeats the last 15 characters of the previous one

Example: the class demo text was 519 characters long. With a chunk size of 120 and overlap of 15, it produced seven chunks. The first chunk ended at a sentence boundary, and the second began with the last few words of the first, so a sentence cut at a boundary still appears whole somewhere. These values are an example, not a universal rule. Start with 300–800 tokens for prose policies and test on real questions.

Too small	Too large
Chunks lose their context and answers become fragments	Search becomes vague and irrelevant text goes to the model

21. Embeddings, with a toy picture

An embedding turns text into a list of numbers. Texts with similar meaning get numbers that point the same way. Here is a toy example with two dimensions (real embeddings have hundreds or thousands):

Text	Dimension 1	Dimension 2
annual leave rules	0.9	0.2
sick leave policy	0.8	0.3
office dress code	0.1	0.9
query: how does leave work	0.85	0.25

The query points almost the same way as the two leave texts and away from the dress code. **Cosine similarity** measures how closely two vectors point the same way. The same embedding model must be used for chunks and queries, or the numbers are not comparable.

22. Vector stores: the options to know

Option	Good for	Think about
Local store (library or file-based)	Demos, small private projects	Backups, sharing, scale
Hosted vector database	Shared production systems	Cost, data location, access control

Database with a vector extension	Teams already using that database	Index tuning and performance

Every stored chunk should carry its text, its vector, and metadata such as source name, page, date and access group. Metadata lets you filter before you search.

23. Retrieval tuning: top-K and filters

- **Top-K:** the number of chunks you pass to the model. Too few misses the answer; too many adds noise and cost. Test values such as 3, 5 and 8 on your test set.
- **Metadata filters:** restrict search to the right document type, date or access group before ranking.
- **Re-ranking (advanced):** a second model sorts the top candidates more carefully. It adds cost and latency, so add it only when measurements show it helps.

24. A sample context block and augmented prompt

```
[System]
You are an HR assistant. Answer ONLY from the context.
If the context does not contain the answer, reply: "I don't know. Please contact HR."
Cite the source name in brackets after each claim.

[Context]
[Source: Annual leave rules, p.2] Employees receive 18 days of annual leave per
calendar year.
[Source: Leave policy, p.1] Sick leave requires notice to the manager on the same
day.

[Question]
How does our leave policy work?
```

Notice the three rules: the instruction comes first, every chunk carries its source, and the question comes last. Test the prompt with an unanswerable question; the correct reply is the abstention sentence.

25. Practice questions (answers follow)

1. A chatbot must answer from last week's policy document. RAG or fine-tuning?
2. Your answers quote facts that are not in the retrieved chunks. Which part do you fix first?
3. Name the four stages of RAG in order.
4. Why must a Hindi question and Hindi chunks use a multilingual embedding model?

5. What does a top-K of 2 risk?
6. Why store the source name as metadata?

Answers: 1. RAG, because the facts change and must be searchable with sources; fine-tuning would be out of date after each change. 2. The augmentation instruction and the retrieval test, because the model should answer only from context. 3. Indexing, retrieval, augmentation, generation. 4. So that Hindi questions and Hindi text land close together in the vector space. 5. The right chunk may be left out, so the answer is incomplete. 6. So the answer can cite where each claim came from and so a reviewer can check it.

26. Full glossary

Term	Plain meaning
RAG	Retrieve relevant text first, then answer with it in the prompt.
Chunk	A piece of a document sized for search.
Chunk overlap	Text repeated between neighbouring chunks.
Embedding	A list of numbers representing meaning.
Cosine similarity	How closely two vectors point the same way.
Vector store	A database of chunks, vectors and metadata.
Top K	The number of best chunks passed to the model.
Augmentation	Building the prompt from instructions, context and question.
Grounding	Whether an answer is supported by the retrieved text.
Abstention	Saying "I don't know" when the context lacks the answer.
Few-shot	Examples in the prompt that teach a pattern.
Fine-tuning	Retraining the model's weights on your examples.

27. Full checklist

Item	Done
Chose RAG, fine-tuning or in-context learning with a reason	[ ]
Chunk size and overlap written down and tested	[ ]
Same embedding model for chunks and queries	[ ]
Metadata stored with every chunk	[ ]

Top-K tested on a 20-question set	[ ]
Augmentation instruction includes abstention and citations	[ ]
Index refresh schedule logged	[ ]

Skills, Subagents and AI Agents

Three words get mixed up constantly: a **skill**, a **subagent** and an **AI agent**. This guide defines each one, shows how to decide which to build, and walks through examples from six industries. It also covers the case most teams miss: **neither**. Every example is self-contained and you do not need the class video.

1. The three definitions in one sentence each

Term	One-sentence definition	Analogy
Skill	A saved set of instructions, examples and sometimes scripts that the model loads when a task matches, so you do not retype them.	A recipe card in your kitchen drawer.
Subagent	A helper model run inside a larger task, with its own separate context, usually for one focused job, and it returns a result to the main conversation.	A specialist you hand one question to, who comes back with an answer.
AI agent	A model in a loop that decides its next step, calls tools (search, read a file, send an email), looks at the result, and repeats until the goal is met or it must stop.	A junior employee who can take actions and must be supervised.

2. The key difference

- **A skill is knowledge you reuse.** It does not act on its own. You, or the model, trigger it, and it shapes the answer. It costs tokens only when loaded.
- **A subagent is a worker you delegate to.** It runs a separate task with its own context window, so the main conversation stays clean. It is useful when a job would fill the main context.
- **An agent is a decision-maker that acts.** It chooses actions, calls tools, and loops. It is the most powerful and the most risky, because each step can change something in the real world.

> **Rule of thumb:** if it only needs the right instructions, it is a **skill**. If it needs to do one bounded job on its own, it is a **subagent**. If it must choose its own steps and use tools toward a goal, it is an **agent**. If none of those fits, it is **neither**, and a plain prompt or a normal script is the better answer.

3. The decision flow

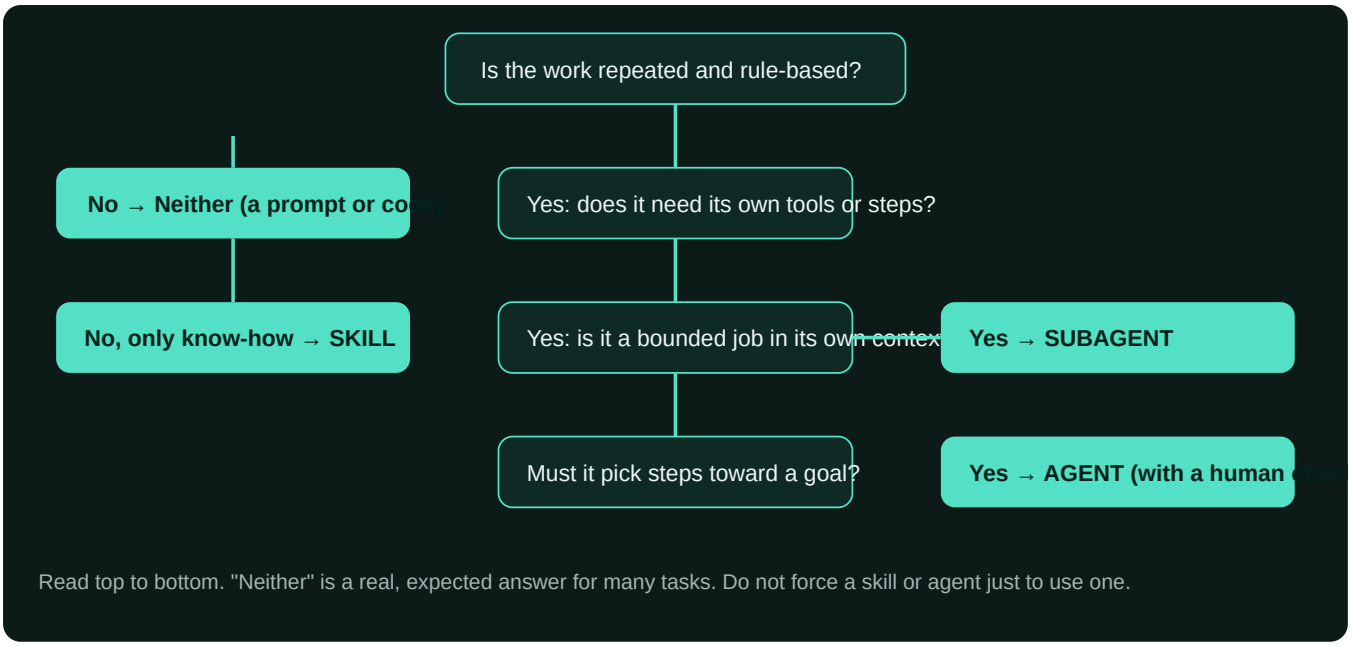


Figure 1. Decision flow. Start with the simplest option and move right or down only when you must.

4. How to create a skill (step by step)

A skill is a folder with one main file, `SKILL.md`. The file starts with a short header that says what the skill is for and when to use it, followed by the instructions.

1. **Pick one repeated task.** Example: "write a product requirements document in our house style". If you do it weekly, it is a skill candidate.
2. **Write the header.** Name, one-line description, and when to use it. The description is how the model decides to load it, so be specific.
3. **Write the instructions.** Role, steps, the required output sections in order, the tone, and what to avoid.
4. **Add two or three examples.** One good input and its ideal output. Examples do more than long rules.
5. **Add a checklist.** The final check before the answer is returned, such as "all required sections present, no invented numbers".
6. **Test on five real inputs**, including one difficult case. Fix the instructions, not the output, so the fix repeats.
7. **Save it** in your skills area (for Claude, under the skills settings) and invoke it by name or by matching request.

```
---
name: prd-copilot
description: Helps a product manager draft a PRD section by section. Use when the
```

```
user asks for a PRD, requirements or product spec.
---
# Role
You are a PRD co-pilot. Draft one section at a time, ask clarifying questions where
depth is missing, and flag gaps before the document is finished.

# Sections, in this order
1. Problem statement (3-5 sentences, no bullet points)
2. Pain points (each tied to a user segment and evidence)
3. Goals and success metrics
4. Scope and non-goals
5. User flow
6. Requirements
7. Risks and open questions

# Checklist before you answer
- Every section present and in order
- No invented numbers; mark assumptions clearly
```

The skill runs only when needed, so it costs tokens on those requests only. It replaces the long prompt you would otherwise paste every time, which is the main token saving.

5. How to create a subagent

A subagent is a named helper with its own description, instructions and allowed tools. Create one when the job is self-contained and would clutter the main conversation, for example "read 40 pages of contracts and list every clause that mentions liability".

1. **Name it for the job**, for example `contract-liability-reviewer`.
2. **Write what it receives and what it returns.** Input: one contract. Output: a table with clause, page and risk level.
3. **Limit its tools.** A reviewer needs read access only. Do not give it write or send permissions.
4. **Ask the main conversation to delegate** that job, and to use only the returned table.

6. How to create an AI agent

An agent needs four things: a goal, a set of tools, a decision rule, and a stopping rule. This is the most important part of the guide.

1. **Define the goal in one sentence**, with a measurable finish, for example "reply to a support ticket with an approved answer, or hand it to a human within 2 minutes".
2. **List the tools, with the least permission each one needs.** Read the knowledge base: yes. Send an email: only after human approval.

3. **Write the decision rule.** "If the ticket mentions a refund over the policy limit, do not answer; route to a human."
4. **Write the stopping rule.** Maximum steps (for example 6), maximum spend, and what counts as done.
5. **Add a human approval gate** before any action that is hard to undo: sending money, sending external messages, changing access, deleting data.
6. **Log every step** (input, tool called, result, decision) so you can explain and debug each run.
7. **Test with fake data first,** then a dedicated test account, then a limited live pilot.

> **Safety for agents:** never give an agent your personal email, your main drive or real customer data while you are learning. Use a test account, minimum permissions and approval before anything external.

7. Worked examples across six industries

Each example shows the right choice and why. The labels "Skill", "Subagent", "Agent" and "Neither" follow the decision flow.

Example A: Retail (e-commerce product descriptions)

Task	Choice	Why
Write a product description in the brand voice	Skill	Same tone and format every time; no tools needed.
Check stock and price, then update the listing	Agent	Chooses steps, reads inventory, writes to the store, needs approval before publishing.
Count the words in a description	Neither	A simple script does this exactly and cheaply.

Example B: Banking and fintech (loan document checks)

Task	Choice	Why
Explain a loan term in plain language	Skill	Fixed style and definitions; no action taken.
Read 80 pages of a loan file and list missing documents	Subagent	A bounded job that would fill the main context.
Chase missing documents, update the file and notify the customer	Agent (human approval before the message)	Multi-step, uses tools, changes records and contacts customers.
Calculate EMI from a rate and tenure	Neither	An exact formula belongs in code, not in a model.

Example C: Healthcare (appointment support)

Task	Choice	Why
Draft a patient-friendly reminder in a fixed, approved style	Skill	Repeated wording, reviewed once by staff.
Summarise a 30-page discharge summary into a checklist	Subagent	Single focused job; reads, returns a table.
Book, move or cancel appointments	Agent (staff approval for cancellations)	Chooses slots and changes a live calendar.
Decide whether a symptom is urgent	Neither (route to a clinician)	Clinical judgement must stay with a qualified person; an AI must not decide alone.

Example D: Education (course and assessment)

Task	Choice	Why
Generate quiz questions in the course format	Skill	Same format each time; the quiz generator skill is one example.
Grade 200 short answers against a rubric	Subagent (one per batch)	Bounded batches keep each context small.
Build a personal study plan from a learner's progress	Agent	Reads progress, picks next lessons, schedules reminders.
Assign a final grade	Neither (teacher decides)	High-stakes judgement belongs to a person.

Example E: Logistics (shipment exceptions)

Task	Choice	Why
Write the standard delay message in the company tone	Skill	Fixed template and tone.
Read carrier tracking logs and find the failed scan	Subagent	Bounded read-and-report job.
Detect a late shipment, reroute it and notify the buyer	Agent (human approves rerouting cost)	Multi-step with money and external messages.
Compute the distance between two warehouses	Neither	A map or distance API is exact.

Example F: Public sector (citizen service requests)

Task	Choice	Why
Explain an application form in simple language	Skill	Approved wording; no action.
Read a request and list which department it belongs to	Subagent	One clear classification job.
Route, log and send status updates across departments	Agent (officer approval for decisions)	Tools, records and communication; a person owns each decision.
Decide eligibility for a benefit	Neither (rule engine plus officer)	Eligibility rules must be auditable and deterministic.

8. Build a skill and an agent: a full walkthrough

The Support Ops case below is the exercise to practise on. Use it to build one skill and one agent, each working on its own.

> **The problem.** A SaaS company handles support through one shared inbox. Five capabilities are needed: (1) read, categorise and route new tickets within minutes, escalating outages and legal or security keywords immediately; (2) make every outbound reply match the company tone and never state a refund or price that is not in the current policy; (3) for an at-risk customer, combine billing, usage logs and past tickets into a one-page briefing before a retention call; (4) grade a weekly sample of closed tickets against a quality rubric and summarise recurring failure patterns; (5) when a customer reports a bug, check the changelog for a known issue and, if none, write it up for engineering with repro steps.

Step 1: map each capability

Capability	Choice	Justification
1. Categorise and route tickets	Agent	Chooses a team from several tools and must escalate immediately; the action (routing) is repeated and changes the workflow. Keep escalation rules in code where possible.
2. Tone-matched, policy-safe replies	Skill (with a check)	Same tone and rules each time; no action taken. The policy check should read the current policy file, not a memorised number.
3. At-risk account briefing	Subagent	A bounded job across three systems that would fill the main context; returns one page.

4. Weekly quality grading	Subagent (batch) plus a skill for the rubric	Batch grading is bounded; the rubric is reusable knowledge.
5. Bug triage against the changelog	Agent, human review before engineering	Searches the changelog, decides whether it is new, and writes a report; a person checks before filing.

Note: no capability needed "neither" here, but step 2's policy number and step 1's escalation rule should be code or data, not model memory. That is the "neither" part of the design.

Step 2: build the skill (reply tone and policy)

```
---
name: support-reply
description: Drafts a customer support reply in the company tone. Use for any
outbound reply to a customer.
---
# Role
You write support replies for a SaaS company. Warm, direct, no jargon.

# Rules
- Never state a refund, discount or price that is not in the policy text provided in
the request.
- If the policy text is missing, write "I'll check this and get back to you" and flag
it.
- Maximum 120 words. One clear next step at the end.

# Examples
Input: "Can I get a refund for last month?" + policy: 14-day refunds
Output: "Thanks for reaching out. Refunds are available within 14 days of purchase,
and yours falls outside that window, so I can't offer one. I'd be glad to help you
with an upgrade or a credit instead."

# Checklist
- Policy figure matches the provided policy text
- No invented numbers
- Under 120 words
```

Test it: give it five real tickets, two with the policy text missing, one asking for a price that is not in the policy. Pass means: no invented figure and a flag where information is missing.

Step 3: build the agent (ticket routing)

1. **Goal:** route each new ticket to one team within 5 minutes, or escalate at once.
2. **Tools:** read the ticket (read only); read the team list (read only); write the route to the ticketing tool (write, but only the route field). No email access.

3. **Decision rule:** if the ticket contains an outage or a legal or security keyword from the escalation list, escalate immediately to the on-call person and stop. Otherwise choose the team that matches the topic.
4. **Stop rule:** a maximum of 4 steps. If no team fits with confidence above the threshold, route to "triage" and flag for a person.
5. **Approval:** escalations notify a person; routes are written automatically only within the rules.
6. **Log:** ticket id, keywords found, team chosen, confidence, and the reason in one sentence.

```
Agent: ticket-router
Goal: route each new ticket to exactly one team, or escalate.
Tools (least permission):
  - read_ticket        (read)
  - read_team_list     (read)
  - set_route_field    (write: route only)
Escalation keywords (checked first, in code): outage, down, data breach, legal, GDPR,
lawsuit
Stop: 4 steps maximum. Unsure → route = triage, flag = needs_human
Log: ticket_id, keywords, team, confidence, reason
```

Test it: send one normal ticket, one with the word "outage", one ambiguous ticket, and one with a broken tool. Pass means: the outage ticket escalates, the ambiguous one goes to triage, and the broken tool stops the run cleanly.

Step 4: check the skill and agent on their own

Each one is tested alone. Do not connect them into a pipeline for this exercise. The skill must produce correct replies on its own; the agent must route correctly on its own, with only the tools it needs.

9. Skills and agents in other tools

Tool	Reusable instructions	Delegated helper	Acting agent
Claude (chat, Desktop and Code)	Skills (SKILL.md) and project instructions	Subagents in Claude Code	Agents with tools, via Claude Code or an agent framework
Lovable	Project knowledge and reusable prompts	Not a separate concept	Connected workflows (for example to n8n) do the acting
ChatGPT	Custom GPTs and Projects with instructions and files	Varies by product	Actions and connected tools, with approval steps

Feature names change often. Check the current documentation for the tool you use before you rely on a name.

10. Common mistakes

- **Making an agent for a fixed template.** If the steps never change, a skill or code is cheaper and safer.
- **Giving an agent write access to everything.** Start read-only, and add one write permission at a time.
- **No stop rule.** An agent with no step limit can loop and spend.
- **Hiding the rules in one long prompt.** Split them: skill for know-how, code for exact rules, agent for choosing steps.
- **Skipping the "neither" answer.** Many tasks need only a prompt or a script.

11. Quick check

1. A team writes the same sales email in the company tone every week. What is it?
2. A model reads 60 invoices and returns a table of mismatches. What is it?
3. A model decides which of three systems to query, then sends a message. What is it?
4. Calculating tax from a fixed table. What is it?

Answers: 1. Skill. 2. Subagent. 3. Agent (with approval before sending). 4. Neither; use code.

12. Glossary and further reading

Term	Plain meaning
SKILL.md	The main file of a skill: header plus instructions.
Subagent	A focused helper with its own context that returns a result.
Agent	A model that chooses steps and uses tools toward a goal.
Tool	An action the agent can call, such as read a file or send a message.
Stop rule	The limit (steps, spend, time) that ends an agent run.
Human-in-the-loop	A person approves an action before it happens.

- Anthropic documentation on agents, tools and skills: docs.anthropic.com.
- Claude Code and skills documentation, with the current feature list; check before you build.

13. Extension: skills, projects and MCP in practice

Exercise A: write and test a skill (30 minutes)

1. Pick one task you repeat every week. Write the header (name, one-line description, when to use it).
2. Write the instructions in this order: role, rules, output sections, checklist.
3. Add two examples: one good input and its ideal output.
4. Test with five inputs, including one hard case and one missing-information case.
5. **Pass when:** every output has the required sections, no invented facts, and the missing-information case is flagged.
6. Version it: write "v1" at the top, and record each change with the reason.

Exercise B: project instructions

A project keeps standing instructions and reference files for one piece of work. Example instruction for a learning project: *"Tone: warm and direct. Audience: beginners. Always define a term before using it. Never invent prices. End each answer with one next step."* Test it with the same question asked two ways; the answers should share the tone and rules.

Exercise C: a safe MCP check

1. List the tool you want an agent to use, and what it can read or write.
2. Connect it to a test account with fake data only.
3. Ask the agent to read one record. Then ask it to write one test record.
4. **Pass when:** the read works, the write needs your approval, and each call appears in the log.

14. Diagram: choosing the right tool for the job

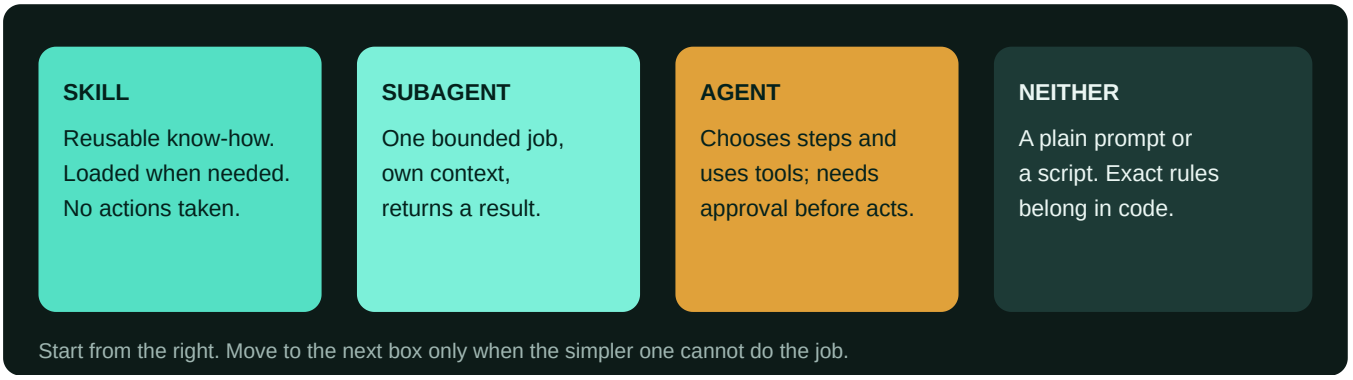


Figure 6. The four choices side by side.

15. Advanced: MCP safety checklist

Check	Pass when
Trusted server	You know who publishes it and read its permissions
Least privilege	Read-only unless a write is required for the task
Data scope	Only the test account and fake data are reachable
Approval boundary	Sends, deletes and access changes need a human
Audit log	Every tool call is recorded without personal data

16. Extended answer key and checklist

Item	Done
Skill written, tested on five inputs, versioned	[ ]
Project instructions tested with two phrasings	[ ]
Agent tools listed with least permission	[ ]
MCP safety checklist completed	[ ]

17. Create a skill from scratch: the full build

This walkthrough builds a skill called `meeting-notes-to-actions`. It turns meeting notes into owned, dated action items. It uses fake notes only.

1. **Define the job in one sentence:** "Turn messy meeting notes into a table of actions with an owner, a due date and a status."
2. **Define inputs and outputs.** Input: notes as plain text. Output: a table with columns action, owner, due date, status (open, done, reassigned, dropped), and a list of open questions.
3. **Write the header.** Name, one-line description, and the words a user would say to trigger it.
4. **Write the rules.** Never invent an owner or date; write "owner needed" instead. Use only the four statuses. Keep each action to one line.
5. **Add two examples.** One complete input and output, and one input with missing owners to show the "owner needed" rule.
6. **Add a checklist** the model runs before answering: every action has a status; no invented names; open questions listed.
7. **Test on five inputs:** a clean set, a messy set, a set with no owners, a set with a date in the past, and a set with a duplicate action.

8. **Version it.** Write "v1" in the header and note every change with the reason and the date.

```
---
name: meeting-notes-to-actions
description: Turns meeting notes into an action table with owner, due date and
status. Use when the user asks for action items, follow-ups or a summary of
commitments from notes.
version: 1
---
# Role
You convert meeting notes into owned, dated action items.

# Rules
- Never invent an owner or due date. Write "owner needed" or "date needed".
- Status is one of: open, done, reassigned, dropped.
- One action per row. Merge duplicates and say so.

# Output
1. Action table: action | owner | due date | status
2. Open questions (bullet list)

# Checklist before answering
- Every row has a status
- No invented names or dates
- Duplicates merged
```

18. Reuse, version and share

Practice	Why it matters
Keep the header accurate	The header decides when a skill is used
Version every change	You can see when an output changed and why
Keep examples current	Examples teach the format better than long rules
Store fake test cases with the skill	Re-testing after a change takes minutes
Remove the skill when unused	Unused skills still add context cost in some tools

19. Project instructions: examples

Project	Instruction (example)	What it prevents
Learning content	"Warm, direct tone. Beginner audience. Define each term before use. Never invent prices."	Jargon and invented figures

Customer replies	"Under 120 words. Never state a refund that is not in the policy text provided."	Unsupported promises
Internal reports	"Summary first. Each figure has a source line. Mark estimates clearly."	Unsourced numbers

Test a project instruction by asking the same question two ways. The answers should share the tone and the rules. If they differ, the instruction is too vague.

20. Claude.ai, Claude Desktop and Claude Code

Surface	Where you use it	Typical use	Skills and projects
Claude.ai	The web chat	Writing, analysis, quick questions	Projects with instructions and files
Claude Desktop	The desktop app	The same chat, with desktop features and connectors	Skills and projects, depending on your plan and settings
Claude Code	A terminal, editor or web session on a code project	Reading and changing code, running tests	Skills, subagents, and CLAUDE.md standing notes

Feature availability depends on plan and version. Check the current help pages before you plan around a feature.

21. Comparable concepts in other tools (and where they differ)

Idea	Claude	Lovable	ChatGPT	Difference to watch
Standing instructions	Project instructions, CLAUDE.md	Project knowledge and reusable prompts	Custom instructions, Projects, custom GPTs	Scope and sharing differ by product
Reusable workflow	Skill	Saved prompt or template in the project	Custom GPT	Skills can include scripts; check each product
Delegated helper	Subagent (Claude Code)	Not a separate concept	Varies by product	Do not assume equivalence
Acting on other apps	Agents with tools, via Claude Code or	Connected services	Actions and connected tools	Approval rules differ

	connectors			

The honest mapping is approximate. Use the product's own documentation before building a process that depends on one feature.

22. Decision drill: skill, subagent, agent or neither

Task	Choice	Reason
Write a weekly update in the company tone	Skill	Repeated know-how
Read 50 pages and list every penalty clause	Subagent	One bounded job that would fill the main context
Find a stuck ticket, reassign it and notify the owner	Agent (approval before notifying)	Chooses steps and acts
Add two prices	Neither	Exact arithmetic belongs in code

23. MCP safety in practice

1. **List each tool** the agent could call, with what it reads and writes.
2. **Connect a test account** that holds only fake data.
3. **Start read-only.** Ask the agent to read one record and confirm the result.
4. **Add a write only when needed,** and make it require approval.
5. **Review the audit log** after each session. Every call should appear.

24. Full quiz (answers follow)

1. A team reuses the same sales email format every week. What is it?
2. A subagent reads 80 invoices. Why not do it in the main chat?
3. Why should an agent have read-only access first?
4. What must every skill header contain?
5. Name the three surfaces discussed for Claude.

Answers: 1. A skill. 2. It would fill the main context and clutter later answers. 3. So a mistake cannot change data, and each added right is a deliberate choice. 4. A name and a one-line description of when to use it. 5. Claude.ai, Claude Desktop and Claude Code.

25. Full checklist

Item	Done
Skill written with header, rules, examples and checklist	[ ]
Skill tested on five inputs, including a hard case	[ ]
Version recorded	[ ]
Project instructions tested with two phrasings	[ ]
Agent tools listed with least permission	[ ]
MCP safety steps followed	[ ]

SkillNavora GenAI Launchpad · Written by Varun Kumar · Part 4 of the automation series · Uses fake data only

n8n Multi-Agent Onboarding: From Zero to Hosted

You will build a small new-hire onboarding system. An HR form sends a webhook to n8n. A Coordinator agent records the hire in a Google Sheet, asks an Access Agent to plan system access, and asks a Scheduling Agent to plan meetings. Everything uses **fake data** and a **dedicated test account**. Follow the steps in order, and check the "You should see" line after each one.

> **Safety first.** Never connect your personal Gmail, your main Google Drive or real employee data to a practice workflow. Create a test Google account and a test sheet. Email sending stays off until a person approves it.

1. The big picture

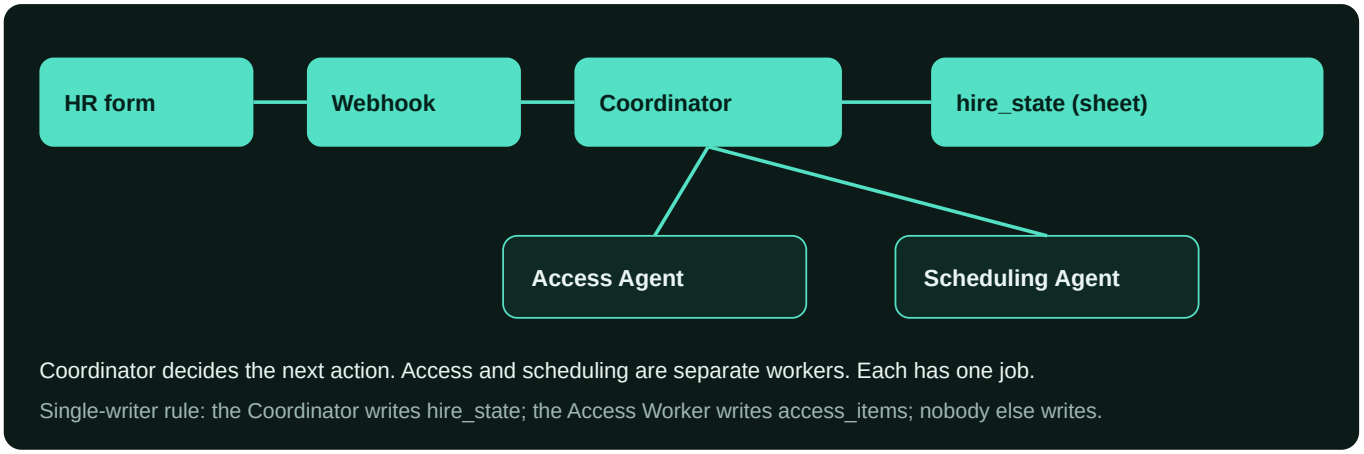


Figure 1. The onboarding system. Arrows show who calls whom.

2. Hosting n8n (choose one path)

n8n can run in three places. Pick one and keep the rest of the guide the same.

Option	Best for	What you manage
n8n Cloud (managed)	Learning, quick start	Nothing on the server; pay per plan
Self-hosted on a VPS (virtual server)	Full control, one fixed monthly server cost	Server updates, backups, HTTPS certificate
Local with Docker	Testing on your own computer only	Your computer must stay on; webhooks need a public address to receive real events

Self-hosting on a VPS: step by step

1. **Choose a VPS plan.** Hosting companies such as Hostinger sell VPS plans. Compare the monthly price, the RAM, and whether a one-click n8n template is included. Prices change; read the live cart before you pay. Hostinger's cart in the class showed a KVM 1 plan with a free domain for the first year, which is a promotion, not a permanent price.
2. **Point a domain or subdomain** (for example `n8n.example.com`) at the server's IP address.
3. **Install n8n** with the one-click template or with Docker. Set an encryption key and a strong admin password.
4. **Turn on HTTPS** so webhook URLs are secure.
5. **Back up** the n8n data folder and the encryption key separately.

You should see: the n8n login page at your domain, over HTTPS, and you can sign in to the editor.

3. Your first workflow: a Webhook test

A **webhook** is a doorbell. The workflow waits until something rings it. Instead of checking repeatedly for new hires, n8n waits, and the HR form sends a POST request when a hire is confirmed.

1. Click **Create workflow**, then add a **Webhook** node.
2. Set **HTTP Method** to POST, **Authentication** to None (for this practice), and **Respond** to Immediately.
3. Copy the **Test URL**. It looks like `https://your-n8n.example.com/webhook-test/<long-id>`.
4. Click **Listen for test event**.
5. In another tab, send a test submission (step 4 below shows the body to use), or paste this into a tool such as an HTTP client:

```
POST https://your-n8n.example.com/webhook-test/<long-id>
Content-Type: application/json

{
  "hire_id": "H-2026-0001",
  "candidate_name": "Test Candidate",
  "department": "Product",
  "role": "PM",
  "manager": { "name": "Test Manager", "email": "manager.test@example.com" },
  "start_date": "2026-09-01",
  "candidate_email": "candidate.test@example.com"
}
```

You should see: the Webhook node shows one item. Open the **JSON** view. Your fields sit under `body`. The `hire_id` is the field that identifies the person in every later step.

> **Test versus production.** The *test* URL works only while "Listen for test event" is open. The *production* URL works all the time, but only after you activate the workflow. Use the test URL while building, then switch the form to the production URL and activate.

Reading the webhook output

Part	What it holds	Use it for
headers	Who sent it, content type	Checking the request came from the form
params, query	URL parameters	Usually empty for a form POST
body	Your data	Everything your workflow needs
webhookUrl, executionMode	Which URL and mode ran	Confirming test versus production

Use the **Schema**, **Table** and **JSON** tabs in the output panel. The JSON view is the most reliable when you are learning.

4. The data contract

A **data contract** is the agreed list of fields and their meaning. Every part of the system uses the same names. Changing a name in one place breaks the others.

Field	Example	Rule
hire_id	H-2026-0001	Unique per hire. The primary key. Never reused.
candidate_name	Test Candidate	Required
candidate_email	candidate.test@example.com	Personal email; required until the work email exists
department, role	Product, PM	Must match a value in the access registry
manager.name, manager.email	Test Manager	Required
start_date	2026-09-01	Must be in the future
run_id	RUN-H-2026-0001-001	Generated by the Coordinator, one per attempt

5. The workbook: four tabs, one writer each

Create a Google Sheet named **onboarding_agents_workbook** using a test Google account. Each tab has **one writer**. This stops two parts of the system from overwriting each other.

Tab	Sole writer	Purpose
hire_state	Coordinator	One row per hire: overall, access and scheduling status
access_items	Access Worker	One row per hire per access item
meetings	Scheduling Worker (not built here)	One row per proposed meeting
access_registry	Hand-maintained (nobody writes it at runtime)	Which apps each role gets, and whether approval is needed

Status words

Tab and column	Allowed values	Terminal?
Overall, access and scheduling status	in_progress, complete, needs_attention	complete and needs_attention end the flow
access_items status	not_requested, requested, awaiting_approval, granted, not_granted	granted and not_granted are final
meetings status	pending, proposed, conflict, confirmed, not_scheduled	confirmed and not_scheduled are final

CSV templates (copy these into the tabs)

Use these headers exactly. Lookups are exact matches, so spelling and capital letters must match.

```
hire_state
hire_id,candidate_name,candidate_email,department,role,manager_name,manager_email,start_date,run_id,overall_status,access_status,sch

access_items
hire_id,run_id,app,level,status,approval_needed,updated_at

meetings
hire_id,run_id,role,meeting_name,duration_minutes,within_days,priority,status,proposed_time,updated_at

access_registry
row_key,app,level,role,drive_role,grant_mode,depends_on
Identity|create|PM,Identity,create,PM,,approval_required,
Identity|create|Designer,Identity,create,Designer,,approval_required,
Identity|create|Developer,Identity,create,Developer,,auto,
Slack|edit|PM,Slack,edit,PM,,auto,Identity|create
Slack|edit|Designer,Slack,edit,Designer,,approval_required,Identity|create
Slack|edit|Developer,Slack,edit,Developer,,auto,Identity|create
Jira|view|PM,Jira,view,PM,,auto,Identity|create
Jira|admin|Developer,Jira,admin,Developer,,auto,Identity|create
Figma|edit|Designer,Figma,edit,Designer,,approval_required,Identity|create
Onboarding Doc|view|PM,Onboarding Doc,view,PM,reader,approval_required,
Onboarding Doc|edit|Designer,Onboarding Doc,edit,Designer,writer,approval_required,
Onboarding Doc|view|Developer,Onboarding Doc,view,Developer,reader,auto,

meetings_template (the reference list, keep it in access_registry or a separate tab)
role,meeting_name,duration_minutes,within_days,priority,purpose
PM,Manager 1:1,60,2,1,Align on 30-60-90 day plan and immediate priorities
PM,Product Vision Walkthrough,45,5,2,Walk through product strategy, roadmap and OKRs
PM,Stakeholder Introduction,30,7,3,Meet key cross-functional stakeholders
Designer,Manager 1:1,60,2,1,Align on design principles and current projects
Designer,Design System Walkthrough,45,5,2,Walkthrough of the design system and tools
Developer,Manager 1:1,60,2,1,Align on engineering standards and sprint cadence
Developer,Tech Stack Walkthrough,45,5,2,Deep dive into the tech stack and dev environment
```

> **Open decisions.** The class workbook flagged one cell to decide before going live: the Developer grant mode for the Onboarding Doc. The sample above uses `auto` for read-only access; confirm your own policy. Delete any example rows before using the sheet for real.

Create the sheet

1. Create the spreadsheet and rename the first tab to `hire_state`.
2. Add the other tabs and paste each CSV header row.
3. Paste the registry rows into `access_registry`.

4. Share the sheet only with your test account.

6. The Google Sheets connection

1. In n8n, open **Credentials** and add a **Google Sheets** credential. Sign in with the test account and approve the access. Give only Sheets access.
2. Add a Google Sheets node, choose **Append Row**, and select the document and tab.
3. Map each field from the webhook body to a column. Use expressions such as `{{ $json.body.hire_id }}`.

You should see: after one test submission, a new row appears in `hire_state`. If it does not, check the document name, the tab name and the credential.

7. The Coordinator agent

The Coordinator is an AI Agent node with a chat model (Anthropic in the class), an **Update row** Google Sheets tool, and two tools that call the Access and Scheduling agents. Its job is to decide the next action, not to do the work itself.

Tool descriptions matter

The agent reads each tool's description to decide when to use it. Write it as a rule: "Update the hire_state row for a given hire_id. Use this to set overall_status, access_status and scheduling_status." Vague descriptions cause wrong calls.

Use the narrowest operation

Google Sheets offers Append, Update, Append or Update, Clear, Create and Delete. Give the agent **Update Row** only. Never give it Clear or Delete. Least permission is the rule.

System message (the instructions)

```
Role: You are the Coordinator for new-hire onboarding.
Steps:
1. Read the hire from the input. Confirm hire_id is present; if not, stop and set needs_attention.
2. Call the Access Worker with these named fields, each separately:
   hire_id, candidate_name, role, manager_name, manager_email, start_date, candidate_email,
   run_id (format RUN-{hire_id}-001).
   Do not join the fields into one string.
3. Call the Scheduling Worker with hire_id, candidate_name, role and start_date.
4. Update the hire_state row with the statuses you received.
Output format (JSON only):
{ "next_action": "dispatch | wait | close", "reasoning": "one sentence" }
Never send email. Never change access directly.
```

Why this helps: named fields prevent one long string that the workers cannot parse. Asking for JSON output makes the next node reliable.

Turn on **Return Intermediate Steps** while building, so you can see each tool call. Turn it off in production if the output gets long.

8. The Access Agent (a separate workflow)

The Access Agent is its own workflow, started by the Coordinator through an **Execute Workflow** or "Call workflow" tool. It receives the named fields from step 7.

1. **Start:** "When Executed by Another Workflow", with the input fields listed (run_id, hire_id, role, manager_name, manager_email, start_date, candidate_email, dispatched_at).
2. **Idempotency check:** a Get Row(s) step reads `access_items` filtered by `hire_id = {{ $json.hire_id }}`. If rows already exist for this hire, stop. This stops the same hire being processed twice.
3. **If node:** true branch means "already processed, stop"; false branch continues.
4. **AI Agent:** reads the role from the input, and uses the **Registry Lookup tool** (read `access_registry`) to find each app for that role.
5. **For each app:** if `grant_mode` is `auto`, write a row with status `requested`. If `approval_required`, write `awaiting_approval` and do not grant.
6. **Write Row tool:** appends to `access_items` only. Never writes to `hire_state`.

7. **Gmail (optional):** send a message only after a person approves. For practice, leave the Gmail tool disconnected.

You should see: running the same hire twice writes rows only once. The second run stops at the If node.

9. The Scheduling Agent (concept only in this build)

The Scheduling Agent reads `meetings`, proposes times for each role's meetings within the `within_days` window, and writes `proposed` rows. It must never confirm a meeting without a person. In this guide, build the Coordinator and Access Agent first; add scheduling once those work.

10. Duplicate prevention and error paths

Risk	Prevention
The same form submitted twice	Check `hire_id` exists before appending a new hire row.
Missing email or hire_id	Validate first; set `needs_attention` and stop.
Wrong role	Look up the role in the registry; if not found, stop with "needs_attention".
Webhook delivery fails	Check the executions list; re-send the test event; use a retry only for safe steps.
Access granted without approval	Rows with `approval_required` stay `awaiting_approval`.

Add an **Error Trigger** workflow that writes "needs_attention" and the error message to `hire_state`, so failures are visible in the sheet, not hidden in logs.

11. Monitor the workflow

- Open **Executions** for each run. A green run may still have made a wrong decision, so read the output of each node.
- Track the **failure rate**. One failure in one production run is a 100 percent failure rate, so judge trends over many runs.
- Track **tokens per run** from the agent's log, and compare with the cost guide in this series.
- Track **run time**. Multi-agent runs take seconds to minutes; set a timeout.

12. The HR Portal: intake form and dashboard

The HR Portal is a small web app (built in Lovable in the class). It has two screens.

Screen 1: New Hire Intake

Section	Fields	Rules
Candidate	Candidate name, candidate personal email	Name required; email must look like an email
Assignment	Department, role, manager	Department and role come from a list; manager from a list
Timing	Start date	Must be in the future
Action	Confirm hire	Generates the hire_id, then sends the webhook POST

Screen 2: Dashboard

A table of hires with columns: hire_id, name, role, start date, overall status, access status and scheduling status. Colour the status (in_progress, complete, needs_attention). Add a "Show webhook status" or "Export as CSV" button if you want operational help.

Build it in Lovable (outline)

1. Create a project and describe the two screens in plain English: "An HR portal with an intake form and a dashboard, using fake data only."
2. Ask for the form fields exactly as in the table above.
3. When the form is ready, paste the **test** webhook URL and ask: "Send the form data to this URL as JSON after Confirm hire."
4. Submit one fake hire and check that it appears in the n8n test panel.

5. Switch to the **production** URL, and activate the workflow in n8n.

> **Do not use real personal data** in a demo app. Fake names and test addresses only, and do not publish a live URL that accepts real submissions.

13. Build order (do not skip steps)

1. Host n8n and sign in.
2. Webhook test: send one fake body and read the JSON.
3. Write one row to `hire_state`.
4. Add the Coordinator with the system message, using a test model key.
5. Build the Access Agent, test duplicates, then connect it.
6. Add the form and the dashboard.
7. Switch to production URLs and activate.

14. Test checklist

Test	Pass when
Valid hire	One row in hire_state, access rows written
Same hire submitted twice	No duplicate access rows
Missing email	needs_attention, no access granted
Unknown role	needs_attention with a reason
Approval-required app	awaiting_approval, nothing granted
Webhook down	Failure visible in executions and in the sheet

15. Quick check

1. Why is `hire_id` the most important field?
2. Which tab does the Access Worker write to, and which does it never write?
3. Why should the Coordinator never be given the Delete operation?
4. What does the If node in the Access Agent prevent?

Answers: 1. It identifies the hire in every step and prevents duplicates. 2. Writes `access_items`; never `hire_state`. 3. Least permission; a mistake could erase the workbook. 4. Processing the same hire twice.

16. Glossary

Term	Plain meaning
Webhook	A URL that starts a workflow when something posts to it.
Test versus production URL	Test works only while you are listening; production works once the workflow is active.
Data contract	The agreed field names and meanings.
Single writer	Only one part of the system may write to each tab.
Idempotency	Running the same request twice gives the same result, with no duplicate rows.
Registry	The reference table of apps and approval rules per role.

- n8n documentation: docs.n8n.io (webhook and Google Sheets node pages).
- Your hosting provider's VPS documentation for installing Docker and HTTPS.

Part 2. Every n8n option you need, explained

This part is a reference. You can read it once, then come back when you meet a node you do not recognise. Node names follow n8n's usual labels; if a label differs slightly in your version, look for the closest match.

2.1 Words you will see on the canvas

Word	Meaning	Example
Workflow	The whole automation: a chain of nodes.	Onboarding workflow
Node	One step. Each node does one job.	Google Sheets: Append row
Trigger	The node that starts the workflow.	Webhook, Schedule, Manual
Connection (wire)	The line that passes data from one node to the next.	Webhook → Sheets
Item	One piece of data flowing through a wire. A node runs once per item unless told otherwise.	One hire = one item
Execution	One complete run of the workflow, with its inputs and outputs saved.	Run for hire H-2026-0001
Credential	A saved, encrypted login for an app.	Google Sheets account (test)
Expression	A formula inside a field, written between double curly braces.	`{{ $json.body.hire_id }}`
Sub-node (tool, model, memory)	A node that plugs into an AI Agent from below.	Anthropic Chat Model, Google Sheets tool

2.2 Trigger nodes: how a workflow starts

Trigger	Starts when	Use it for	Watch out
Manual trigger	You press Execute workflow	Building and testing	Not automatic; never leave it as the only trigger in production
Webhook	An app sends an HTTP request to the node's URL	Forms, apps, other systems pushing events	Use the test URL while listening; the production URL needs the workflow active
Schedule	At a time or interval you set	Daily reports, periodic checks	Time zone; each run is an execution that costs resources
Google Sheets trigger (polling)	New or changed rows appear	Reacting to new spreadsheet rows	Polls on an interval, so delays are normal
Execute Workflow Trigger ("When Executed by Another Workflow")	Another workflow calls it	Sub-workflows such as the Access Agent	Pass the fields you need explicitly
Error Trigger	Another workflow fails	Central error handling and alerts	Set it up in a separate workflow and link it in settings

2.3 Action and logic nodes you will use most

Node	What it does	Key setting
Google Sheets	Read rows (Get row(s)), add rows (Append row), change rows (Update row), upsert (Append or Update)	Document and sheet selection; the Operation dropdown
HTTP Request	Calls any web API (GET, POST, PUT, DELETE)	URL, method, headers, body; use a credential for keys
If	Splits items into true and false branches	Condition: field, operator, value
Switch	Routes items to many branches by value	Rules per branch; add a fallback output
Merge	Joins two branches back together	Mode: append, combine by key, or choose input
Set (Edit Fields)	Adds, renames or removes fields	Keep only the fields the next node needs
Code	Runs JavaScript or Python on the items	Use for exact rules (dates, validation); keep it small
Filter	Keeps only items that match a condition	Condition builder
Wait	Pauses for a time or until a webhook resumes it	Use it for approval waits
Loop Over Items (Split in Batches)	Processes items in groups	Batch size
Gmail	Sends or reads email	Use a test account; keep sending behind approval
Slack, Notion, and other app nodes	Post messages or change app records	Test channel or test workspace only

Stop and Error	Ends a branch on purpose with a reason	Message text for the log
Sticky note	A comment on the canvas	Document each section for future you

2.4 AI nodes

Node	Role	Notes
AI Agent	The decision-maker: reads its input, uses tools, loops until done	Needs a chat model; tools are optional but each one widens risk
Chat Model (Anthropic, OpenAI, Google, Groq and others)	The model the agent talks to	Choose by speed, quality and cost (tokens guide)
Memory	Remembers earlier turns in the same conversation	Off for stateless processing; on for chat
Tool (sub-node)	An action the agent may choose	Give each tool a clear description the agent can read
Output Parser	Forces a structured output such as JSON	Prefer structured output for anything another node reads
Basic LLM Chain	One model call with no tool loop	Use it when no decision-making is needed (cheaper and more predictable)

> **Choose the simplest node that works.** A fixed template is a Set node. A single model call is a Basic LLM Chain. Only a task that must choose steps and tools needs an AI Agent.

Part 3. Build your own n8n workflow: a 12-step recipe

Use this recipe for any workflow, not only onboarding. Each step has a check. Do not move on until the check passes.

1. **Write the goal in one sentence.** Example: "When HR confirms a hire, record it in the sheet and ask for access planning."
   Check: you can say what "done" looks like.
2. **List the inputs and outputs** with names and types. Use the data contract table from Part 1.
   Check: every input has a name and an example value.
3. **Choose the trigger**: Manual for building, Webhook for real events, Schedule for time-based work.
   Check: you know whether it runs once per event or on a timer.
4. **Add the first action node** (for example Google Sheets: Append row) and connect it to the trigger.
   Check: Execute the node. One row appears in the test sheet.
5. **Map fields with expressions**, for example `{{ $json.body.candidate_name }}`.
   Check: the mapped preview shows the right values, not "undefined".
6. **Add validation** with an If node: required fields present, email shape, start date in the future.
   Check: a bad input takes the false branch and writes "needs attention".
7. **Add the decision or AI step** only if a rule cannot do it. Start with a Basic LLM Chain before an AI Agent.
   Check: the output is structured and parseable.
8. **Add the external action last**, and put it behind an approval (Wait node or a manual step).
   Check: nothing external happens without approval in the test run.
9. **Add an error path**: an If on error, a Stop and Error node, or the Error Trigger workflow.
   Check: force a failure (a wrong sheet name). The failure appears in the log and the sheet shows "needs attention".
10. **Make it idempotent**: check whether the item already exists before writing it.
    Check: running the same input twice leaves one row.
11. **Test with five inputs:** normal, missing field, duplicate, bad date, and a tool failure.
    Check: each result matches the expected column in your test plan.
12. **Switch to production**: copy the production URL, update the form, activate the workflow, and run one fake end-to-end test.
    Check: the production execution shows success and the sheet shows one row.

Part 4. The multi-agent onboarding build, node by node

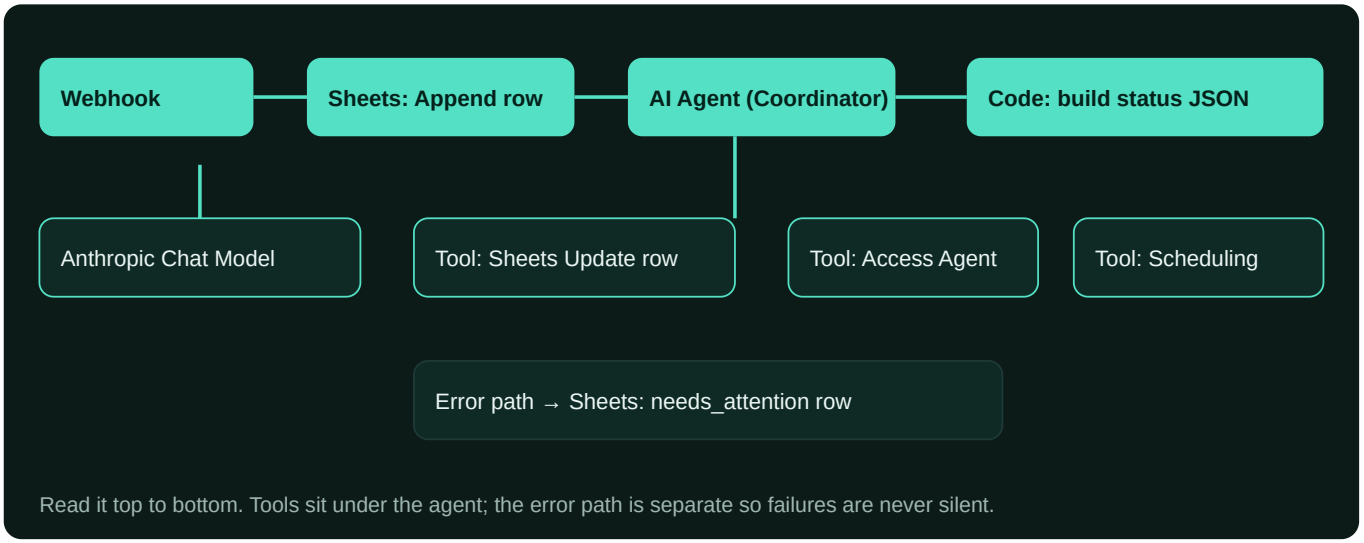


Figure 2. The Coordinator as built in n8n. Sub-nodes plug into the agent from below.

4.1 Coordinator (AI Agent)

1. Add an **AI Agent** after Sheets: Append row.
2. Attach **Anthropic Chat Model** to the Chat Model connector. Choose a model; start with the smallest that passes your test set.
3. Attach the tools: **Google Sheets (Update Row only)**, **Call Access Agent** (Execute Workflow tool), and **Call Scheduling Agent**.
4. Write each tool's description as a rule. Example for Sheets: "Update the hire_state row for a given hire_id. Use it to set overall_status, access_status and scheduling_status."
5. Paste the system message from Part 1 section 7, and leave Memory off.
6. Turn on **Return Intermediate Steps** while building.
7. Run once with the fake hire. Open each tool call in the output and confirm the named fields arrived separately.

4.2 Access Agent (separate workflow)

Node	Setting	Why
When Executed by Another Workflow	Input fields: run_id, hire_id, role, manager_name, manager_email, start_date, candidate_email, dispatched_at	Explicit input; nothing implicit
Sheets: Get row(s)	Document onboarding_agents_workbook, sheet access_items, filter hire_id = {{ $json.hire_id }}	Idempotency: look before you write
If	Condition: number of rows is greater than 0	True means already processed: stop
AI Agent with Registry Lookup tool	Tool reads access_registry for the role	Rules come from the sheet, not from memory
Code (or Set)	Turn each registry row into one access item with status requested or awaiting_approval	Deterministic: approval rules are code, not model choice
Sheets: Append row	Sheet access_items only	Single-writer rule
Gmail (optional, disconnected in practice)	Only after a person approves	No external message without approval

> **Single-writer rule.** The Access Worker writes access_items. The Coordinator writes hire_state. Neither writes the other's tab. When two parts write one table, rows overwrite each other and the status becomes unreliable.

4.3 Scheduling Agent (build after the first two work)

It reads the meetings template, proposes times inside each meeting's within_days window, and writes meetings rows with status proposed. A person confirms. It never sends invitations on its own in this guide.

Part 5. Hosting and operations, done safely

Topic	What to do	Why
Environment variables	Keep the encryption key, database settings and base URL in environment variables, not in workflow nodes	Changing one value changes every workflow
Credentials	Create them in n8n's credential store; use test accounts; least permission	Secrets stay encrypted and out of exported files
HTTPS	Put n8n behind HTTPS with a domain	Webhooks carry data; plain HTTP exposes it
Updates	Update on a schedule; read release notes; test in a copy first	Updates can change node behaviour
Backups	Back up the database and the encryption key separately	Without the key, saved credentials cannot be read
Logs	Review executions daily during testing; keep logs without personal data	Failures must be visible
Access control	One owner account; add test users only when needed	Fewer people can change production
Exports	Export workflows as JSON to version control, without credentials	History and rollback

Part 6. Test plan with pass criteria

Test	Input	Pass when
Happy path	Complete fake hire	One hire row; access rows; status complete or awaiting_approval
Missing field	No hire_id	needs_attention; no access rows
Duplicate hire	Same hire_id twice	One set of access rows
Malformed JSON	Body is not JSON	Request rejected or error row written; no crash
Unknown role	role = Analyst	needs_attention with a reason
Failed tool call	Wrong sheet name	Error path runs; sheet shows needs_attention
Timeout	Slow tool (simulate with Wait)	Run stops at the limit and is logged
Retry	Transient failure	Retry twice at most, then needs_attention
Human approval	Approval-required app	awaiting_approval; nothing granted until a person changes it

Part 7. Measure the workflow

Metric	How to calculate	Target to aim for while learning
Success rate	Successful executions ÷ all executions	Rising toward 100 percent on the test set
Failure rate	Failed executions ÷ all executions	Falling; investigate every failure
Duplicate rate	Duplicate hire rows ÷ hire rows	Zero
Time to complete	Median time from webhook to hire_state complete	Seconds for a test run
Escalation rate	needs_attention rows ÷ hire rows	Known and explained, not zero by force
Cost per run	Model tokens per run × price (tokens guide)	Recorded for every test batch

Part 8. Extended quick check and answer key

1. Which trigger starts a workflow when HR submits a form?
2. Where should a fixed approval rule live: in a prompt or in a registry and code?
3. Why put an If node before writing the access items?
4. A test run shows a green node but a wrong status. What do you check first?

Answers: 1. Webhook. 2. In the registry and code, so it does not depend on the model's memory. 3. It stops duplicates. 4. The node's output: the actual values it passed on, not just its colour.

Part 9. Final checklist before you go live

Item	Done
Test accounts only; no personal Gmail or Drive	[ ]
Fake data only in every test	[ ]
Test URL worked; production URL is set in the form	[ ]
Duplicate check present and tested	[ ]
Error path visible in the sheet	[ ]
Human approval before any external action	[ ]
Credentials stored in n8n, not in nodes or exports	[ ]
Backups of database and encryption key taken	[ ]

Part 10. Labs: five practice exercises with fake data

Lab 1: webhook in, row out (30 minutes)

1. Create a workflow with a Webhook node (POST, test URL) and a Google Sheets Append row node to a test sheet.
2. Send the fake hire body from Part 1. Map `hire_id`, `candidate_name` and `department`.
3. **Pass when:** one row appears, with the right values in the right columns, and the execution is green.

Lab 2: validation branch (30 minutes)

1. Add an If node after the webhook. Condition: `hire_id` is not empty AND `candidate_email` contains "@".
2. True branch: Append row to `hire_state`. False branch: Append row to a sheet named `review_queue` with the reason.
3. **Pass when:** a valid body lands in hire_state and an invalid body lands in review_queue, and nothing is lost.

Lab 3: duplicate check (40 minutes)

1. Before the append, add a Google Sheets Get row(s) node filtered by `hire_id`.
2. Add an If node: row count equals 0 goes to the append branch; otherwise go to a "duplicate" branch that writes a note to the review queue.
3. **Pass when:** sending the same body twice leaves one hire row and one review note.

Lab 4: approval wait (40 minutes)

1. Add a Wait node after the access plan is written, set to wait for a webhook resume.
2. Write the approval request to the sheet with status `awaiting_approval`.
3. Send a second test webhook to the resume URL, with the field `approved: true`.
4. **Pass when:** the status changes to `granted (simulated)` only after the second request, and never before.

Lab 5: forced failure and error route (30 minutes)

1. Rename the target sheet to a wrong name, then run the workflow.
2. Connect an error route that appends a row to `review_queue` with the error message.
3. **Pass when:** the failure appears in the executions list, the review queue shows the message, and the workflow does not report success.

Part 11. Troubleshooting table

Symptom	Most likely cause	What to check first
Webhook shows no data	Listening stopped, or wrong URL	Click "Listen for test event" again; confirm test versus production
Field shows "undefined"	Expression points to the wrong path	Look at the JSON view; the data sits under `body`
Sheet row has the wrong column	Mapping to the wrong column header	Compare header names in the sheet with the mapping
Agent calls the wrong tool	Tool description is vague	Rewrite it as a rule with the trigger condition

Agent loops	No stop rule	Set a maximum step count and a clear finish condition
Production run does nothing	Workflow not active	Activate the workflow; check the production URL in the form
Credential error	Expired or wrong account	Reconnect the test account; check the scopes granted
Duplicate rows	No idempotency check	Add Lab 3's check before every append

Part 12. Cost, speed and model choice for workflows

- **Use the cheapest step that works.** Set and If nodes cost nothing extra in model terms; AI nodes cost tokens on every run.
- **Measure per run:** record the model tokens reported by the agent for each test batch, then apply the tokens-guide formula.
- **Reduce input:** pass only the fields the agent needs; never the whole execution history.
- **Cap output:** set a maximum output length for every agent call.
- **Watch run time:** an agent that waits on several tools can take minutes. Set a timeout and log it.

Part 13. Full glossary

Term	Plain meaning
Node	One step in a workflow.
Trigger	The node that starts a workflow.
Execution	One saved run of a workflow.
Credential	A saved, encrypted login for an app.
Expression	A formula inside a field, such as `{{ $json.body.hire_id }}`.
Sub-node	A model, tool or memory node attached to an agent.
Webhook	A URL that starts a workflow when something posts to it.
Idempotency	Running the same input twice gives one result.
Single writer	Only one part of the system writes each table.
Error route	A path that runs when a step fails, so failures are visible.

Part 14. Quiz (answers follow)

1. Which node holds the hire data sent by the form?
2. Where is a fixed approval rule stored, and why?
3. What should happen to a duplicate hire?
4. A production run shows nothing. What do you check first?
5. Why should an AI node have a maximum output length?

Answers: 1. The Webhook node, under `body`. 2. In the registry and code, so it does not depend on the model's memory and can be audited. 3. It is recorded in review and no second set of rows is written. 4. Whether the workflow is active and the production URL is in the form. 5. It caps cost and keeps the answer a predictable length.

Part 15. Build-your-own planning sheet

Question	Your answer
What event starts my workflow?	[ ]
What fields must it contain?	[ ]
What is the final output, and where does it go?	[ ]
Which step needs judgement (AI), and which needs a rule (If, Set, Code)?	[ ]
Which step changes something outside the system, and what approval does it need?	[ ]

What happens on failure, and who is told?	[ ]
How do I prevent duplicates?	[ ]
How will I test it with fake data?	[ ]

Part 16. Industry workflows you can build with the same pattern

The onboarding build uses one pattern: an event starts the workflow, a rule or agent decides, a sheet records state, and a person approves anything external. Use the same pattern for other industries. Each example below uses fake data and a test account.

Industry	Trigger	Rule or agent step	State record	Human approval before
Retail	New product listing submitted by a seller form	Check price and category against the policy sheet; draft a description with a 50-word cap	Listing sheet with status draft, approved, rejected	Publishing the listing
Clinics	Appointment request webhook (fake patient data)	Check slot availability in a calendar sheet; propose two times	Booking sheet with status proposed, confirmed, declined	Any confirmation message to a patient
Logistics	Shipment exception from a tracking feed	Classify the exception (delay, damage, missing) using a fixed list	Exceptions sheet with status open, resolved, escalated	Rerouting or refunds
Education	Quiz submission from a form	Score against a rubric; draft feedback	Results sheet with status scored, reviewed	Sending feedback to a learner
Public service	Service request form	Route to a department from a fixed list	Requests sheet with status routed, acknowledged, closed	Any decision on eligibility (never automated here)

> **Eligibility, diagnosis and pricing decisions stay with a qualified person.** Automation may prepare the information; it must not make the decision.

16.1 Retail example, node by node

1. **Webhook** (POST): receives the listing form, test data only.
2. **If**: price is greater than zero and category is in the list. False goes to a review sheet.
3. **Sheets: Get row(s)** on the policy sheet for the category rules.
4. **AI Agent** (or a Basic LLM Chain) drafts a description under the word cap. Output is structured: title, description, word count.
5. **Code**: checks the word count and removes any price or claim from the draft.
6. **Sheets: Append row** to the listing sheet with status draft.
7. **Wait** for approval (a second webhook from the reviewer's test form).
8. **Sheets: Update row** to status approved. Publishing to a real store is out of scope for this lab.

Pass when: the draft never includes a price, the word count is under the cap, and nothing reaches status approved without the second webhook.

Part 17. Production readiness, in depth

Area	Question to answer	Evidence you need
Reliability	What happens when a tool is down?	Test with the tool disconnected; error route runs
Data protection	Which personal fields are stored, and for how long?	Field list with a retention rule
Access	Who can edit the workflow and the sheets?	Named owners; least permission on each credential
Change control	How is a change tested before it goes live?	Test run on fake data; export of the workflow JSON, without credentials
Monitoring	Who sees a failure, and how fast?	Error route writes to a visible sheet; daily review during the first week
Cost	What does one run cost in model tokens?	Recorded tokens per run; the tokens-guide formula
Rollback	How do you return to the last good version?	Saved export of the previous version

17.1 Monitoring routine

- **Daily (first week):** open the executions list; read the output of any red or unusual run; log the cause in the review sheet.

- **Weekly:** count success rate, duplicate rate, escalation rate and median time; compare with last week.
- **After any change:** run the five-input test set again before you trust the workflow.

Part 18. Security checklist for automations

Risk	Control
Credential leaks through exports	Export workflows without credentials; store exports in version control with no secrets
A public webhook accepts anything	Add an authentication check (header or signed token) before production; reject unsigned requests
Prompt injection through form text	Treat form text as data, never as instructions; keep the system message separate; limit agent tools
Over-broad tool access	One permission per tool; read-only where possible
Personal data in logs	Log IDs and statuses, not names or emails
Accidental external messages	No live email or chat node until a human approval step is in place and tested

Part 19. Build a second workflow alone (exit exercise)

1. Choose one industry from Part 16 and write the goal in one sentence.
2. Write the data contract: at least five fields, with one unique ID.
3. Build the trigger, one rule, one state record and one approval step, without an AI agent.
4. Add an AI step only where a rule cannot decide, and cap its output.
5. Run five fake inputs and fill in the test table from Part 6.
6. **Pass when:** the workflow passes the five tests, the duplicate check holds, and the approval step is required before any external action.

Part 21. Credentials and the approval gate, in practice

Credential	Test version	Production rule
Google Sheets	Test Google account, test sheet only	Service account with access to one named sheet
Model provider key	Test key with spending limit	Key stored in n8n credentials; rotated on a schedule
Email (Gmail)	Not connected during build	Connected only after the approval node is tested
Chat (Slack)	Private test channel	Named channel; messages contain IDs, not personal data

21.1 The approval gate, built and tested

1. After the access plan is written, add a **Wait** node set to resume on a webhook.
2. Write the request to the sheet with status `awaiting_approval` and the approver's name.
3. Send a test webhook to the resume URL with `approved: true` from the approver's test form.
4. **Pass when:** the status changes only after the resume call, and a second resume call with `approved: false` writes `not_granted`.

Part 22. Common design mistakes and fixes

Mistake	Why it hurts	Fix
One giant workflow for everything	Hard to test; one change breaks all	Split into sub-workflows with explicit inputs
Agent does arithmetic or date checks	Models make small errors	Use Code or Set nodes for exact rules
Status written by two parts	Rows overwrite each other	Single-writer rule per table
No stop rule on an agent	Loops and spends tokens	Maximum steps and a finish condition
Test data mixed with production	Fake records reach real sheets	Separate sheets and credentials for test and production

Part 23. Frequently asked questions

Question	Answer

Why use n8n instead of only an AI chat?	n8n runs the same steps every time, records state, and waits for approval; a chat does not.
Is an AI Agent always needed?	No. Most steps are rules. Use an agent only where a rule cannot decide.
Why keep a sheet if n8n has executions?	Executions show runs; the sheet shows the business state a team can read and act on.
What if the webhook URL changes?	Update the form to the production URL and activate the workflow again; test first.
How do I stop a runaway agent?	Deactivate the workflow, then fix the step limit and the stop rule before re-enabling.

SkillNavora GenAI Launchpad · Written by Varun Kumar · Part 5 of the automation series

Make.com: Scenarios, Routers and AI Agents

Make is a visual automation tool. You connect apps as boxes on a canvas and draw the flow between them. This guide takes you from your first scenario to the Resume Analysis Agent from the class. It uses fake data and a test account throughout.

1. Words you need first

Word	Meaning
Scenario	One automation: a chain of modules that runs in order.
Module	One step, such as "Watch responses" or "Add a row".
Trigger	The first module, which starts the scenario (a new form response, a schedule, a webhook).
Bundle	One item of data passing between modules.
Router	A split that sends each bundle down one or more paths based on filters.
Filter	A condition on a path, for example "score is above 70".
Connection	The saved login to an app (a credential). Use a test account.
Operation	Spend of the plan's credits per module run. Check the plan page for current credit rules.

2. The Make screen

The left menu has: **Org**, **Scenarios**, **AI Agents**, **Credentials**, **Webhooks**, **MCP Toolboxes**, **Templates**, **Data stores** and **More**. The canvas in the middle is where you build. The bottom toolbar has **Run once**, a schedule toggle (for example every 15 minutes) and the undo and settings icons.

3. Your first scenario: form to sheet

1. Click **Scenarios**, then **Create a new scenario**.
2. Add a **Google Forms** module, choose **Watch Responses**, and connect the test account. Pick a test form with two fields.

3. Add a **Google Sheets** module and choose **Add a Row**. Select a test sheet.
4. Map each form answer to a column using the mapping panel (click a field, choose the answer).
5. Click **Run once**, then submit one test response to the form.

You should see: the module bubbles turn green, the sheet gets a new row, and the output panel shows the values you mapped.

4. Scheduling and settings

- **Schedule:** the toggle sets how often the scenario checks for new items. The class example was every 15 minutes. Shorter intervals use more operations.
- **Sequential running:** keep it on when order matters, so two runs cannot write at the same time.
- **Error handling:** right-click a module to add an error route. Route failures to a sheet row or a message so they are visible.
- **Notifications:** decide when Make emails you about failed runs. Turn on notifications for failures, not for every success.

> **Before you enable a scenario,** check three things: its operations per month against your plan, its notification setting, and what happens on failure. A scenario left running on a schedule keeps spending operations.

5. Routers and filters

A router is a fork in the road. The class example, the Resume Analysis Agent, uses one.

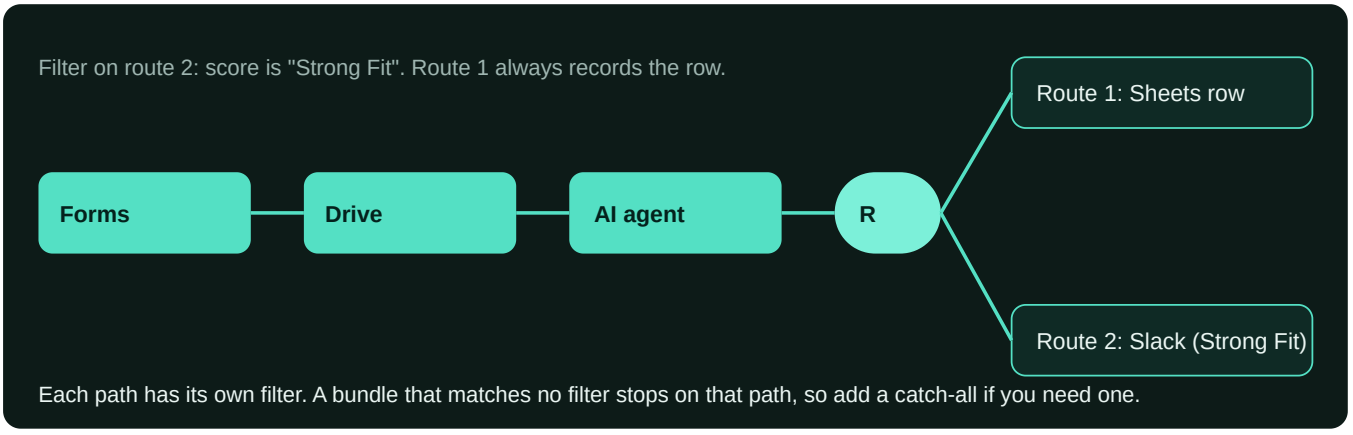


Figure 1. A router splits bundles into paths. Each path has a filter.

1. Click the wrench icon between two modules and choose **Add router**.
2. Add a filter to each route (for example the AI score field equals "Strong Fit").

3. Test with one bundle that matches and one that does not. Confirm only the right route runs.

6. The Resume Analysis Agent (class example)

The class scenario was built to read a candidate's resume and sort the candidate. The full chain is:

1. **Google Forms (Watch Responses):** the candidate submits a form with a resume file link.
2. **Google Drive (Download a File):** the resume file is fetched from Drive.
3. **AI agent:** reads the resume, uses a knowledge source (the job criteria) and returns a fit rating.
4. **Router:** route 1 adds a row to Google Sheets for every candidate; route 2 sends a Slack message only for "Strong Fit" candidates.

The agent in the screenshot also had an **Add tool** option, an **Add MCP** option, a **Chat** option and a knowledge attachment. The AI model module was an OpenAI connection in the class.

> **Privacy.** A resume contains personal data. For practice, use a fake resume with a made-up name and email. Do not send real candidates' files to a test scenario, and do not post Slack messages to a shared channel during testing.

Build checklist

Step	You should see
Forms trigger runs on a test response	One bundle with the form fields
Drive download returns a file	A file name and size in the output
AI step returns a fixed-format rating	One word from your allowed list (for example Strong Fit, Possible, No Fit)
Router sends only matching bundles to Slack	Slack message only for Strong Fit
Failed AI step	Error route writes a row marked "needs attention"

7. Maia: building with a description

Maia is Make's AI assistant (labelled Beta in the class screenshots). You describe what you want in plain English, and Maia proposes modules. The panel offers "Continue where we left off", "Get inspiration" and "Ask a question about Make".

1. Open a scenario and click **Build with Maia**.

2. Describe the goal in one sentence: "When a form is submitted, download the attached file, ask an AI model to rate it, and add a row to a sheet."
3. Review every module Maia proposes. Check each connection, field mapping and filter yourself.
4. Never trust a generated scenario without running it on fake data first.

Maia is a drafting aid. The person who runs the scenario owns its results, so review every step.

8. MCP Toolboxes and AI Agents in Make

MCP (Model Context Protocol) is a standard way for an AI app to use outside tools. Make's MCP Toolboxes group tools that an agent can call. Give an agent only the tools it needs, and read-only tools where possible. The same least-permission rule from the skills and agents guide applies here.

9. Failure, retries and cost

Issue	What to do
Module fails on a bad file	Route the error to a "needs attention" row; fix the input; re-run that bundle.
Scenario runs too often	Lengthen the schedule or use a webhook trigger instead of polling.
Operations run out	Check the plan usage page and reduce the schedule or the number of modules per run.
Duplicate rows	Store the form response ID and skip responses already in the sheet.

Check Make's current plan and credit page before you estimate cost. Plans and credit rules change.

10. Quick check

1. What starts a scenario?
2. What does a filter do on a router path?
3. Why should you use a test connection?
4. What should you do before enabling a scenario that Maia built?

Answers: 1. The trigger module (for example a new form response). 2. Decides whether a bundle goes down that path. 3. So a mistake cannot touch real accounts or data. 4. Run it on fake data and check every module and filter.

11. Glossary and further reading

- Make help centre: help.make.com (scenarios, routers, Maia and MCP pages).
- Make pricing and credit page: check the live page before you estimate operations.

Part 2. Every Make option you need, explained

Make's labels can change between releases. The terms below are the concepts you will meet; if a menu name differs, look for the closest match.

2.1 Words on the canvas

Word	Meaning	Example
Scenario	The whole automation	Resume Analysis Agent
Module	One step (a bubble on the canvas)	Google Sheets: Add a Row
Bundle	One item of data passing between modules	One form response
Route (connection)	The line that carries bundles from one module to the next	Watch Responses → Download a File
Mapping	Choosing which earlier field fills a later field	Email field → Email column
Connection	Saved login for an app	Google account (test)
Data store	Make's built-in table for keeping records between runs	Processed response IDs
Operation	One module action; plans include a set number. Check the live plan page.	Each module run counts

2.2 Trigger modules: how a scenario starts

Trigger type	Starts when	Use it for	Watch out
Schedule	At the interval you set (the class used every 15 minutes)	Checking a sheet or inbox regularly	Each check uses operations even when nothing is new
Watch responses (Google Forms)	A new form response arrives	Intake forms	File uploads come as links; you need a Drive download step
Custom webhook	Another app posts to the Make URL	Real-time events from your own app	Use a test run while building; activate the scenario for live use

Watch new rows (Sheets)	A new row appears	Reacting to sheet data	Sheet changes can be batched; test with a few rows
Manual run	You press Run once	Building and debugging	Not automatic

2.3 Module types you will use

Module type	Examples	What to set
App action	Google Drive: Download a File; Google Sheets: Add a Row; Slack: Send a Message; Gmail: Send an Email	Connection, the action, and the field mapping
Flow control	Router, Filter, Repeater, Iterator, Aggregator, Sleep	Routes, filter conditions, the batch logic
Data tools	Set variable, Text parser, JSON parse, Tools	Use these to clean data before an AI step
Data store	Data store: Add record, Search records, Get record	Store, key field, record fields
AI module	An AI model call (OpenAI in the class example; check the apps list for other providers)	Connection, model, prompt, and output format
AI agent	The agent builder: tools, knowledge, chat	Goal, tools with least permission, stop rule
Error handler	Break, Resume, Commit, Rollback, Ignore directives on a module	Choose per module (section 4)

2.4 Filters and routers in detail

Concept	Plain meaning	Example
Router	Splits one route into several; each path has a filter	Path 1: all candidates to Sheets; Path 2: Strong Fit to Slack
Filter	A condition on a path; only matching bundles pass	AI rating equals "Strong Fit"
Fallback path	A path with no filter that catches everything else	Route anything unclassified to a review sheet
Text operators	Contains, equals, starts with	Rating text contains "Strong"
Number and date operators	Greater than, less than, after	Start date after today

> **Always add a fallback path** on a router. Without it, a bundle that matches no filter stops quietly and you will not know it happened.

Part 3. Build your own Make scenario: a 12-step recipe

1. **Write the goal in one sentence.** Example: "When a candidate submits a fake resume, rate it and log it."
   Check: you can state the output.
2. **List the inputs and fields**, with example values.
   Check: every field has a name and a sample.
3. **Choose the trigger**: manual to build, a form or webhook for real events, a schedule for polling.
   Check: Run once returns one bundle.
4. **Add the first action** and map a field. Run once.
   Check: the target row or message shows the mapped values.
5. **Add validation**: a filter that stops empty or malformed bundles, routed to a review path.
   Check: a bad input goes to review, not to the output.
6. **Add the AI step last among the data steps.** Ask for one fixed word from a list, not free text.
   Check: the output matches the list every time in five tests.
7. **Add the router** with a filter per path and a fallback.
   Check: each test bundle reaches exactly one path.
8. **Add the external action** (Slack, Gmail) only on the right path and only with a test channel or address.
   Check: the test channel receives one message per matching bundle.
9. **Set error handling** on each risky module (section 4).
   Check: force a failure; the error route writes "needs attention".
10. **Prevent duplicates** with a data store: search for the response ID before writing.
    Check: running the same bundle twice writes one row.
11. **Test five inputs**: normal, empty field, duplicate, unexpected AI output, and a tool failure.
    Check: each result matches the test plan (section 6).
12. **Activate and watch the first runs**, then review history daily while learning.
    Check: the run history shows green runs and any red runs are explained.

Part 4. Error handling, explained simply

Directive	What it does	Use it when
Resume	Continues the scenario using a substitute value	A non-critical field is missing

Ignore	Skips the failed bundle and continues the run	Losing this one item is acceptable and logged
Break	Stores the failed bundle to retry later	A temporary outage (for example an app is down)
Commit	Keeps the work done so far and stops the run	Partial results are still valid
Rollback	Undoes changes where the app supports it	An all-or-nothing change

For practice, use **Break** for external steps (so you can retry them) and **Resume** for optional fields. Write the choice in a note on the scenario, so the next person knows why it is set that way.

Part 5. The Resume Analysis Agent, built step by step (fake data only)

> **Fake data only.** Create a test Google Form with a made-up candidate (for example "Test Candidate", candidate.test@example.com) and a sample file. Do not upload real resumes. This tool must not make hiring decisions; a person reviews every result.

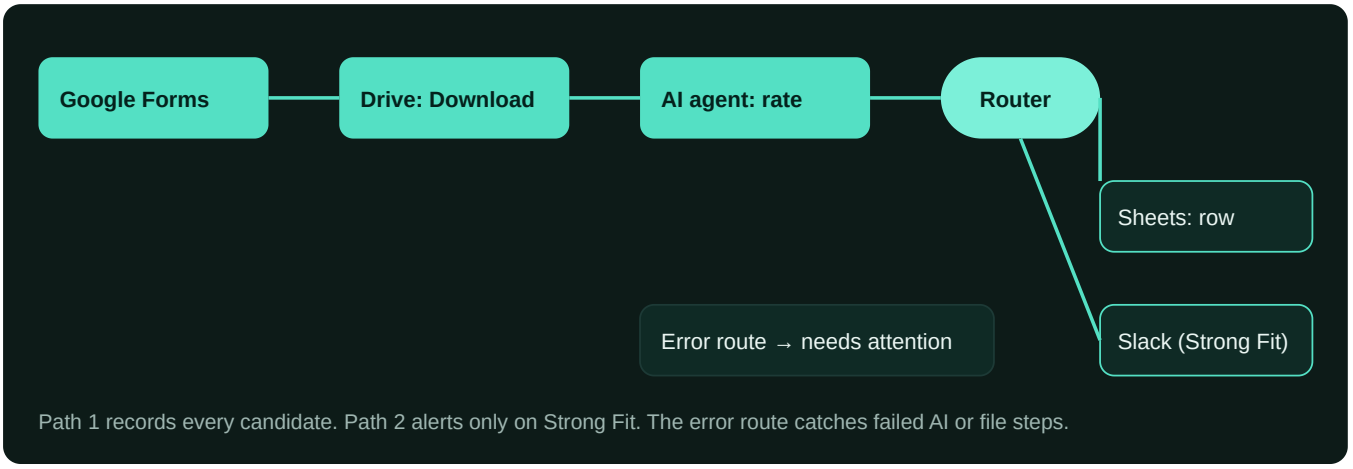


Figure 3. The Resume Analysis Agent layout, redrawn for SkillNavora.

1. **Create the scenario** "Resume analysis (test)" and add **Google Forms: Watch Responses**. Choose the test form and run once after one fake submission.
2. **Add Google Drive: Download a File**. Map the file ID from the form's upload field. Run once. *Check:* the bundle now includes the file name and content.
3. **Add the AI step.** Use one prompt with a fixed output: "Reply with exactly one of: Strong Fit, Possible, No Fit. Base the answer only on the job criteria provided." Provide the criteria as fixed text, not a real job list.
   Check: five fake resumes return only those three words.

4. **Add the router.** Path 1: Google Sheets Add a Row for every bundle (fallback-free). Path 2: a filter "rating equals Strong Fit", then Slack Send a Message to a test channel.
   Check: a Strong Fit test reaches both paths; a No Fit test reaches only Sheets.
5. **Add error handling.** On the Drive and AI modules, set Break; add an error route that writes "needs attention" and the error text to a sheet.
   Check: disconnect the Drive link for one run; the error row appears and no Slack message is sent.
6. **Add a duplicate check** with a data store keyed by form response ID.
   Check: the same response submitted twice writes one row.
7. **Review every output** before any real use. A person decides what happens next.

Part 6. Maia, the AI scenario builder

Maia (labelled Beta in the class screenshots) drafts a scenario from a plain-language description. The panel offers "Continue where we left off", "Get inspiration" and "Ask a question about Make".

Maia can help with	You must still check
Drafting the module order from a description	Each connection and which account it uses
Suggesting modules you have not used before	Every field mapping and filter
Explaining an existing scenario	Whether the scenario is safe to run with real data
Answering "how do I" questions about Make	That the answer matches the current product (check the help centre)

Rule: never run a Maia-built scenario on real data before you have run it on fake data and read each module's output.

Part 7. MCP and Make

MCP (Model Context Protocol) is a standard way for AI applications to use outside tools. Make's MCP toolboxes group tools an agent can call. A tool is a single action, such as reading a sheet or posting a message.

Concept	Plain meaning	Safe default
Input	What the agent sends to the tool	Only the fields the tool needs
Output	What the tool returns to the agent	Read-only results where possible

Scope	Which account data the tool can reach	A test account with fake data
Permission	Whether the tool can read, write or send	Read first; add write only when needed
Approval boundary	The point where a person must approve	Before any send, delete or access change
Audit log	A record of each tool call	Keep it, without personal data

Part 8. Full test plan

Test	Input	Pass when
Happy path	Fake strong candidate	Sheets row and Slack alert
Weak candidate	Fake no-fit candidate	Sheets row only
Unexpected AI output	Forced odd answer	Error route, no Slack message
Missing file	Form with no upload	Error route with a clear reason
Duplicate response	Same response twice	One row only
Slack down	Disconnect Slack	Break directive stores the bundle for retry

Part 9. Common errors

Error	Cause	Fix
Empty output field	Mapping points to the wrong module	Re-map from the latest module output
Connection expired	Login was revoked	Reconnect the test account
Scenario runs but nothing happens	Router has no matching path and no fallback	Add a fallback path
Operations used up quickly	Short schedule on a large scenario	Lengthen the schedule; reduce modules
Duplicate rows	No duplicate check	Add the data store check (Part 3, step 10)

Part 10. Final checklist before you go live

Item	Done

Test account and fake data only	[ ]
Every router has a fallback path	[ ]
Error routes on Break or Resume for every risky module	[ ]
Duplicate check tested	[ ]
Human review before any hiring-related action	[ ]
Operations and schedule checked against the live plan page	[ ]
Run history reviewed after the first ten runs	[ ]

Part 11. Labs: five practice exercises with fake data

Lab 1: form to sheet (25 minutes)

1. Create a test Google Form with two fields. Build the scenario: Watch Responses, then Add a Row.
2. Submit one fake response and run once.
3. **Pass when:** one row appears with each answer in the right column.

Lab 2: filter and fallback (30 minutes)

1. Add a router after the form. Path 1 with a filter "answer equals Yes" goes to Sheets. Path 2 is a fallback that goes to a review sheet.
2. Submit one "Yes" and one "No" response.
3. **Pass when:** each response reaches exactly one path, and nothing disappears.

Lab 3: data store duplicate check (40 minutes)

1. Create a data store with a key field for the response ID.
2. Before the Sheets module, search the store for the response ID; continue only if nothing is found, then add the record.
3. **Pass when:** submitting the same response twice writes one row.

Lab 4: AI output lock (40 minutes)

1. Add an AI module with a prompt that asks for one word from a list (Strong Fit, Possible, No Fit).
2. Add a filter that accepts only those three words; send anything else to the review path.
3. Run five fake inputs, including one that tempts a long answer.
4. **Pass when:** every bundle reaches a path, and no free-text answer reaches the sheet.

Lab 5: break and retry (30 minutes)

1. Set Break on the Slack module. Disconnect the test Slack connection for one run.
2. Reconnect and run the incomplete execution from the scenario's history.
3. **Pass when:** the bundle is retried and the message arrives once, not twice.

Part 12. Troubleshooting table

Symptom	Most likely cause	What to check first
Module shows "no data"	Upstream module produced no bundle	Run the previous module once and read its output
Mapping shows a stale value	Old output cached	Re-run from the module that changed
Router sends nothing	No matching filter and no fallback	Add the fallback path
Scenario is on but not running	Schedule off, or trigger not set	Check the schedule toggle and the trigger settings
Connection needs re-authorisation	Login expired or permissions changed	Reconnect the test account
Operations run out early	Frequent schedule or many modules	Lengthen the schedule; check the usage page
AI output varies	Free-text answer	Ask for one word from a list; lock with a filter

Part 13. Choosing between Make, n8n and a custom app

Situation	Good fit	Why
Visual, fast automation across many business apps	Make	Visual routers and filters; quick to test
Self-hosting, custom code and data control	n8n	Can run on your own server with code nodes
Core product logic with users and a database	A custom app (for example on Replit)	Full control of the user experience and data

The three can work together. For example, a form in an app, an automation in Make, and an agent in n8n. Keep each boundary clear and document it.

Part 14. Full glossary

Term	Plain meaning
Scenario	One automation made of connected modules.
Module	One step, such as an app action or a flow control.
Bundle	One item of data moving through a scenario.
Router	A split into paths, each with a filter.
Filter	A condition that decides whether a bundle continues.
Fallback path	A path that catches anything no filter matched.
Data store	A built-in table that keeps records between runs.
Directive	An error rule on a module: Resume, Ignore, Break, Commit or Rollback.
MCP	A standard way for AI apps to use outside tools.

Part 15. Quiz (answers follow)

1. What starts a scenario?
2. Why add a fallback path to a router?
3. How do you stop duplicate rows?
4. Why lock AI output to a list of words?
5. What should a Break directive do with a failed bundle?

Answers: 1. The trigger module, such as a new form response. 2. So unmatched bundles are not lost. 3. Search a data store for the response ID before writing. 4. So the filter can route predictably and no free text reaches the sheet. 5. Store it so it can be retried.

Part 16. Build-your-own planning sheet

Question	Your answer
What starts my scenario?	[ ]
Which apps do I connect, and with which test account?	[ ]
Where does each path end?	[ ]
Which module is the AI step, and what is its locked output?	[ ]
Which step is external, and who approves it?	[ ]

What is the error directive on each risky module?	[ ]
How do I prevent duplicates?	[ ]

Part 17. Industry scenarios for Make

Industry	Trigger	Flow control	Record	Approval before
Retail	New order from a test store form	Router: standard vs. high-value orders	Orders data store	Refund or change to an order
Recruiting (fake candidates)	Application form response	Router with fallback: Strong Fit, Possible, No Fit, Needs review	Candidates sheet	Any message to a candidate
Clinics	Appointment request (fake data)	Filter on slot availability	Booking sheet	Any confirmation to a patient
Education	Quiz submission	Filter on score band	Results sheet	Sending feedback
Logistics	Tracking exception	Router by exception type	Exceptions sheet	Rerouting or refund

> **Hiring, medical and eligibility decisions must not be automated here.** The scenario may sort and prepare, but a person decides.

17.1 Recruiting example, node by node (fake candidates)

1. **Google Forms: Watch Responses** (test form, fake candidate).
2. **Data store: Search records** by response ID. If found, stop (duplicate).
3. **Google Drive: Download a File** (fake file).
4. **AI module** returns one word from the fixed list, plus a short reason field of 20 words or fewer.
5. **Router**: Path 1 Strong Fit; Path 2 Possible; Path 3 No Fit; Path 4 fallback to Needs review.
6. **Each path** writes one row to the candidates sheet with the path name.
7. **Strong Fit only**: Slack message to a test channel that says "Review needed" and gives the row number, not the candidate's details.
8. **Data store: Add record** with the response ID, so the duplicate check works next time.

Pass when: every fake response reaches one path, the duplicate is stopped, and the Slack message contains no personal details.

Part 18. Operations and cost, without vendor prices

Every module run uses operations, and an AI call also uses model tokens. Plan both before you enable a scenario. Use the live plan page for the numbers; this table shows how to count.

Item	How to count	Example (fake numbers)
Operations per bundle	One per module that runs	8 modules = 8 operations per bundle
Bundles per month	Submissions per day times 30	20 per day = 600 per month
Operations per month	Bundles times operations per bundle	600 × 8 = 4,800
Polling cost	Runs per day from the schedule that find nothing	Every 15 minutes = 96 runs per day, even when empty
Model tokens	Input plus output per AI call, times calls	Use the tokens-guide formula with current rates

Lesson: a frequent schedule on a large scenario spends operations on empty checks. A webhook trigger avoids that cost when the event is real-time.

Part 19. Production checklist

Item	Done
Every router has a fallback path	[ ]
Error directive set on each external module	[ ]
Duplicate check tested with the same input twice	[ ]
Test connections only during build; production connection reviewed	[ ]
Operations per month estimated from the live plan page	[ ]
Notifications set to failures only	[ ]
Run history reviewed after the first ten runs	[ ]
Named owner for the scenario and its data	[ ]

Part 20. Exit exercise

1. Pick one industry from Part 17 and write the goal in one sentence.
2. Build the trigger, one filter, one router with a fallback, and one record step, with fake data.

3. Add the AI step last, with a locked one-word output.
4. Run five fake inputs: normal, empty, duplicate, odd AI output, and a failed module.
5. **Pass when:** every input reaches the right path, the duplicate is stopped, and the failure is visible.

Replit: Build, Run, Share and Deploy an AI App

Replit is a browser-based workspace where you write code, run it, and publish it on the web, with no installation on your computer. This guide goes from your first running page to an AI-powered app with secure keys, saved data, and a deployed address. Plans and prices are deliberately not covered here; check Replit's own pricing page when you need them.

> **Running example.** A small "Dish Description Studio": a form takes a dish name, category and price, calls an AI model, and shows a short description. It is the same food-seller example used in the tokens guide.

Part A. Basics

1. Words you need

Word	Meaning
Repl	A project: its files, its running app and its settings.
Workspace	The browser editor: file tree, code editor, console and preview.
Console	Where the program prints messages and errors.
Preview	The running app shown inside the workspace.
Secret	A private value (such as an API key) stored outside the code.
Deployment	A published copy of the app at a public address.
Replit Agent	An AI that builds or edits the project from your written request.
Environment variable	A named setting the program reads at runtime, such as `AI_API_KEY`.

2. Create an account and your first project

1. Sign up at replit.com. For practice, use a test account, not your work or personal main account.
2. Click **Create Repl**. Choose a template for a web app in a language you know (for example Python with a web framework, or Node.js).
3. Give the project a clear name, such as `dish-studio-practice`.

4. Press **Run**.

You should see: the console shows the server starting, and the preview pane shows a page. If the preview is blank, look at the console first.

3. Learn the workspace in five minutes

Area	What you do there
File tree (left)	Create, rename and delete files. Keep the main file easy to find.
Editor (centre)	Write code. Save happens automatically in most setups.
Console (bottom)	Read output and errors. Read the first error line, not the last.
Preview (right)	See the running page. Refresh after a change if needed.
Packages panel	Add libraries your code imports.
Secrets panel (lock icon)	Store API keys and private values.
Version history	See earlier saved states and roll back a change.

4. Make one change and see it run

1. Open the main file and change the page heading to `Dish Description Studio`.
2. Save, then refresh the preview.
3. If nothing changes, read the console for a red error line.

Common error: a missing package. Read the error ("No module named ..." or "Cannot find module ..."), add the package in the Packages panel, and run again.

5. Build the input form

Add a form with three fields: dish name, category (a dropdown: Breakfast, Lunch, Dinner, Dessert), and price in rupees. Use a submit button. Give every field a visible label. Labels help screen-reader users and make the form easier to use with a keyboard.

```
<form method="post" action="/describe">
  <label>Dish name <input name="dish" required></label>
  <label>Category
    <select name="category">
      <option>Breakfast</option><option>Lunch</option>
      <option>Dinner</option><option>Dessert</option>
    </select>
  </label>
  <label>Price (Rs) <input name="price" type="number" min="1" required></label>
  <button type="submit">Describe my dish</button>
</form>
```

You should see: the form in the preview. Submitting it does nothing useful yet; that is the next part.

Part B. Intermediate: connect an AI model safely

6. Store the key as a secret

An AI call needs an API key from your model provider. Never put the key in the code, in a screenshot, in a chat, or in a public project.

1. Create a key in your AI provider's console, with a monthly spending limit set there.
2. Open the **Secrets** panel and add `AI_API_KEY` with the key as its value.
3. In code, read it from the environment: Python `os.environ["AI_API_KEY"]`; Node `process.env.AI_API_KEY`.

> **If a key leaks,** delete it in the provider console at once and create a new one. Changing the code does not undo an exposed key.

7. Write the AI call with limits

Every call should have three guards: a fixed instruction, a capped output length, and a logged usage count. These make cost predictable (see the tokens guide).

```
import os
# Use your provider's official SDK. Names below are placeholders.
from ai_client import Client

client = Client(api_key=os.environ["AI_API_KEY"])

SYSTEM = ("You write short, appetising menu descriptions for local food sellers. "
          "Warm tone. Maximum 50 words. No prices or claims about health.")

def describe(dish, category, price):
    user = f"Dish: {dish}\nCategory: {category}\nPrice: Rs {price}"
    reply = client.generate(system=SYSTEM, user=user, max_tokens=120,
temperature=0.7)
    print("tokens in/out:", reply.usage.input_tokens, reply.usage.output_tokens)
    return reply.text
```

You should see: after one submission, the console prints the token counts and the preview shows a description of about 50 words.

8. Handle failure gracefully

- If the key is missing, show "AI is not configured" instead of a crash.

- If the call times out, show a retry message and do not retry in a tight loop.
- If the output is empty, show a fallback message and log the input.

9. Save results to a database

To keep a history of generated descriptions, add a small database. Replit offers a built-in database option in the workspace; alternatives include a hosted database service. Store only the fields you need: dish, category, price, description, time, and the token counts. Do not store personal data for a practice app.

Table: descriptions	Type	Note
id	number	Primary key, automatic
dish, category	text	From the form
price	number	Validated: greater than zero
description	text	From the AI call
input_tokens, output_tokens	number	For cost tracking
created_at	time	Automatic

Part C. Advanced: deploy, monitor and collaborate

10. Deploy the app

1. Open the **Deploy** panel and choose the type that fits: a website or web app (always-on address) is the usual choice for this example. Scheduled or background types suit other jobs.
2. Check the build command and run command the panel shows. They should match how you ran the app.
3. Deploy, then open the public address in a private browser window.
4. Test a normal submission, an empty form, a negative price, and a simulated AI failure.

You should see: the same page as in the workspace, at a public address, with secrets working. Deployment options and limits change; confirm them on Replit's documentation before you rely on them.

11. Logs, errors and rollback

- **Logs:** check the deployment logs after each change. Log the request type, the status and the token counts, never the API key.
- **Rollback:** use version history to return to an earlier saved state if a change breaks the app.

- **Health check:** keep one page that returns "ok" so you can see at a glance that the app is up.

12. Collaboration

Invite a teammate to a project only for practice work with fake data. Agree who edits which file. Use version history and comments to explain changes. Do not share secrets with collaborators; they should add their own key for their own testing.

13. Using the Replit Agent well

1. Describe the app in one paragraph: who uses it, what the inputs are, what the outputs are, and what it must never do (for example: never store customer phone numbers).
2. Ask for one feature at a time, then test it.
3. Read every file the Agent changes. Ask it to explain anything you do not understand.
4. Never let the Agent paste a real key into code. Check the secrets panel yourself.

Part D. Other tools you may need

Tool	Why you need it	Safe practice
AI provider account (console)	Gives the API key for model calls	Set a spending limit; use a test key
GitHub	Stores code history outside Replit, and lets you move projects	Use a practice repository with no secrets in it
A database (built-in or hosted)	Stores generated results	Store only the fields you need
Spreadsheet (for example Google Sheets)	Quick data sink or log for non-technical teammates	Test sheet only
Browser and dev tools	Inspect network calls and errors	Never paste keys into screenshots
Email or messaging (optional)	Notifications to a person	Approval before sending; test account only

Part E. When Replit is the right choice

Choose Replit when	Choose another tool when
You want to build and deploy a web app from a browser	You only need a no-code workflow between apps (use n8n or Make)

You are learning to code and want instant feedback	Your app needs heavy, specialised infrastructure from day one
You need a prototype for users to test quickly	You need strict data residency rules that the platform cannot meet (check first)

Part F. Test checklist

Test	Pass when
App runs in the workspace	No red error in the console
Key comes from secrets only	Code contains no key; the call works
Output is capped	Description stops near 50 words
Missing key	A clear message, no crash
Negative or empty price	Validation message; no database write
Deployed address works	Normal path returns a page
Rollback works	An earlier version restores the page

Part G. Common problems

Problem	Fix
"Module not found"	Add the package in the Packages panel and run again.
Key error at runtime	Check the secret name matches the code exactly.
Works in the workspace, fails when deployed	Check the run command and the port setting in deployment.
Blank preview	Read the console; the first red line is the cause.
Output too long, costs rising	Lower max tokens; ask for a word limit; check usage logs.

Part H. Quick check

1. Where should an API key be stored?
2. What is the first thing to read when the preview is blank?
3. Why cap output length on every AI call?
4. What is a safe way to share a practice project?

Answers: 1. In Secrets, read from the environment. 2. The first red line in the console. 3. It keeps cost and response size predictable. 4. Share only fake data, and never share secrets.

Glossary

Term	Plain meaning
Repl	A project in Replit.
Secret	A private value kept outside the code.
Deployment	A published version at a public address.
Environment variable	A named setting the program reads at runtime.
Version history	Saved earlier states you can return to.

- Replit documentation: docs.replit.com.

Part I. Build a full app, step by step (fake data only)

Step 1: plan the app on one page

Item	Example for the Dish Description Studio
Who uses it	A test seller (fake account)
Inputs	Dish name, category, price
Outputs	A 50-word description and a saved history
Must never	Store real phone numbers, or call an AI without a cap
Done when	A test seller can submit, see the description, and find it in history

Step 2: build the form, then the route

1. Build the form from the basic guide and confirm every field shows a label.
2. Add one route that accepts the form and returns the description (a placeholder reply first).
3. **Pass when:** the placeholder appears in the preview for a valid submission.

Step 3: validate on the server

- Reject an empty dish name and a price that is zero or negative.
- Return a clear message, not a crash, for each rejection.
- **Pass when:** five bad inputs each return a readable message and nothing is saved.

Step 4: connect the AI call with limits

1. Store the key as a secret. Read it from the environment only.
2. Cap output length, set temperature, and log token counts.
3. **Pass when:** the description stops near 50 words, the log shows token counts, and no key appears in code or logs.

Step 5: save history

1. Create the table from the basic guide with only the fields you need.
2. Write one row per successful call. Do not write rows for rejected inputs.
3. **Pass when:** a history page lists saved rows, newest first.

Step 6: test the failure paths

Test	Pass when
Secret removed	"AI is not configured" message, no crash
AI call times out	Retry message; no tight retry loop
Empty model reply	Fallback message; input logged without personal data
Very long dish name	Input capped and explained

Part II. Logs, versions and rollback

- **Logs:** record the route, status, duration and token counts. Never record keys or personal data.
- **Version history:** save a named checkpoint before each risky change.
- **Rollback:** if a change breaks the app, restore the last checkpoint, then find the cause in the console.
- **Health check:** one route that returns "ok" lets you check the app is up at a glance.

Part III. Collaboration and review

1. Invite only a practice teammate, and only to a project with fake data.
2. Agree who edits which file, and write a one-line note on each change.
3. Each person adds their own test key; nobody shares a key.
4. Review every change the Agent makes before you accept it.

Part IV. The Replit Agent, used well

Do	Avoid
Describe the app, the inputs, the outputs and the limits in one paragraph	Asking for the whole app in one request
Ask for one feature, then test it	Accepting large unreviewed changes
Ask it to explain code you do not understand	Pasting a real key into the chat
Check the secrets panel yourself	Trusting that a key was stored safely

Part V. Other tools you will need, and why

Tool	Use	Safe practice
AI provider console	API key and spending limit	Set a limit; use a test key
GitHub	Backup and move projects	No secrets in the repository
Database (built-in or hosted)	Saved history	Store only needed fields
Spreadsheet	A simple log for teammates	Test sheet only
Browser developer tools	Inspect requests and errors	Never share screenshots with keys

Part VI. Full glossary and quiz

Term	Plain meaning
Repl	A project in Replit.
Secret	A private value kept outside the code.
Deployment	A published version at a public address.
Checkpoint	A saved version you can return to.
Route	A web address in your app that does one job.

1. Where should an API key be stored?
2. Why validate on the server as well as in the form?
3. What should a rejected input write to the database?
4. What is the first thing to check when a deployed app fails?

Answers: 1. In Secrets, read from the environment. 2. The form can be bypassed, so the server must check too. 3. Nothing. 4. The logs, and whether the same secret and run command exist there.

Part VII. Four app examples you can build

App	Users	Core feature	Data stored	Risk to watch
Feedback form	Fake customers	Form, validation, a thank-you page	Rating and comment (no names)	Abusive text; add a length limit
Quiz app	Learners	Questions, scoring, a results page	Score and time	Answer key exposed in the page source
Inventory tracker	Shop staff (test)	Add and update stock, low-stock alert	Item, count, updated time	Wrong counts; keep a change log
Dish Description Studio	Sellers (test)	AI description with a cap	Dish, price, description, token counts	Cost growth; cap output and log usage

VII.1 Build the quiz app (fake data, about 60 minutes)

1. **Plan:** five questions, one correct option each, a score out of five.
2. **Data:** store the questions in a list in code for this practice.
3. **Route:** show one question at a time; record each answer in the session.
4. **Score:** compare with the answer key on the server, never in the page.
5. **Results:** show the score and the questions answered wrongly.
6. **Pass when:** a full run gives the correct score, and reloading the page does not reveal the answer key.

VII.2 Build the inventory tracker (about 90 minutes)

1. **Table:** items with id, name, count, updated_at, and a change log table with item id, change, time.
2. **Routes:** list items, add an item, change a count. Every change writes a log row.
3. **Validation:** the count cannot go below zero; the name cannot be empty.
4. **Alert:** items at or below a threshold show a low-stock label.
5. **Pass when:** a change appears in the log, a negative count is rejected, and the low-stock label appears at the threshold.

Part VIII. Architecture in one page

Layer	What it does	Where it lives in Replit
Page	Shows the form and results	The web files in the project
Routes	Receive requests, validate, call the database or the model	The server file
Data	Stores records	The database option you chose
Secrets	Keys and private settings	The Secrets panel
Deployment	The public address and its run command	The Deploy panel

Keep the layers separate. The page never talks to the database directly, and the model key never reaches the page.

Part IX. Deployment and operations checklist

Item	Done
Secrets set in the deployment, not only in the workspace	[ ]
Run command matches what worked in the workspace	[ ]
Health route returns ok	[ ]
Logs checked after the first deployment and after each change	[ ]
Checkpoint saved before each deployment	[ ]
Usage and spending limit checked on the provider and Replit pages	[ ]
Test data removed from any production store	[ ]

Part X. Mobile, accessibility and polish

- Test the deployed address on a phone-width screen; forms must stack to one column.
- Every input has a visible label, and error messages sit next to the field.
- Keyboard users can reach every control in a sensible order.
- Colour is never the only signal for an error or a status.

Part XI. Exit exercise

1. Choose one app from Part VII and write its plan on one page.
2. Build the page, one route with validation, and one stored record, using fake data.
3. Deploy, then test a normal input, an empty input, a bad input and a missing secret.
4. **Pass when:** all four tests give the expected message, and the deployed app matches the workspace.

Part XII. Data, teams and change control

- **Schema first:** write the table list and the fields on one page before you create anything.
- **Migrations:** change a table in one small step, test it on fake data, then deploy; never edit a live table by hand.
- **Backups:** export the data on a schedule you set, and test one restore on a second project.
- **Teams:** one owner per project; reviewers read changes before they reach the deployed app.
- **Branches or checkpoints:** use a checkpoint before each risky change, and name it with the date and the reason.

XII.1 Exit test for the team workflow

1. Create a checkpoint, make a change, deploy to the test address, then roll back to the checkpoint.
2. **Pass when:** the rolled-back app matches the earlier behaviour and the logs show both steps.

Part XIII. Choosing Replit, Lovable or n8n for a job

Job	Best first choice	Why
A web page or app that users use directly	Replit or Lovable	Builds the interface and the server together
Moving data between apps on a schedule or event	n8n or Make	Connectors and routing without writing a server
A custom app that needs a workflow behind it	Replit for the app, n8n for the workflow	Each does the job it is designed for; connect them with a webhook

Check the current feature list for the tool you choose; products change quickly.

Part XIV. Frequently asked questions

Question	Answer
Do I need to install anything?	No. You work in the browser. Install only what your project's packages need, through the project itself.
Where do I keep an API key?	In the Secrets panel, read from the environment. Never in code, screenshots or chat.
Why does it work in the preview but fail when deployed?	Check the deployment's run command, the port setting, and whether the secrets exist in the deployment.
How do I undo a bad change?	Return to the last checkpoint in version history, then find the cause in the console.
Can I share my project?	Yes, with a test teammate and fake data only. Do not share secrets.
Should I use the Agent for everything?	Use it for drafts and small features. Read every change and test it yourself.
What about plans and prices?	Check Replit's live pricing page on the day you decide. This guide does not list prices.

Part XV. Common mistakes and fixes

Mistake	Fix
Hard-coded key in the code	Move it to Secrets and rotate the key at once
No server-side validation	Validate in the route, and return a clear message
No output cap on an AI call	Set max tokens and log the usage
Deploying without testing failure paths	Run the four failure tests in Part VI
Real personal data in a practice app	Replace with fake data and delete the real rows

Part XVI. Ten build recipes, from small to larger

Each recipe is one short build with a pass check. Do them in order; each one teaches a skill the next one needs.

#	Recipe	New skill	Pass when

1	A page that says hello, with your name	Run, preview, edit	Change is visible after refresh
2	A form that echoes the input back	Forms and routes	Submitted value appears on the next page
3	Validation with two rules	Server checks	Bad input returns a clear message
4	A counter stored in memory	State within one run	Count rises by one per click
5	A saved list in a database table	Storing records	Records survive a page reload
6	A secret read from the environment	Secrets	Code contains no key; the app still works
7	One AI call with a 50-word cap	Model calls and limits	Output stops near 50 words; usage is logged
8	A history page of saved AI results	Reading stored data	Newest first; no personal data shown
9	A checkpoint, a change, and a rollback	Version history	The rolled-back app matches the old behaviour
10	A deployed copy at a public address	Deployment and logs	Normal, empty, bad, and missing-secret tests pass

> **Stop rule for a recipe:** if a pass check fails twice, read the first red console line, change one thing, and test again. Do not change several things at once.

Part XVII. Worked walkthrough: Dish Description Studio, recipes 1 to 10

1. **Recipe 1–2:** the page shows the title "Dish Description Studio" and a form with dish, category and price. Submitting shows the same values on the next page.
2. **Recipe 3:** the server rejects an empty dish name and a price of zero, each with a clear message. Nothing is saved for a rejected input.
3. **Recipe 4:** a small counter shows how many descriptions this session has created, so you can see the route runs once per submission.
4. **Recipe 5:** each successful submission writes one row: dish, category, price, time. Reload the page to check the row survived.
5. **Recipe 6:** the model key lives only in Secrets. Remove it for one test to confirm the friendly "not configured" message, then put it back.

6. **Recipe 7:** the AI call uses a fixed system instruction, a 120-token output cap, and logs the token counts. Check that a 50-word description results.
7. **Recipe 8:** the history page lists the newest rows first. It shows dish, category, description and time, with no contact details.
8. **Recipe 9:** save a checkpoint, change the heading, then roll back. The heading returns and the rows remain.
9. **Recipe 10:** deploy, open the public address in a private window, and run the four tests from Part VI on the deployed copy.

Final check: the deployed app should behave the same as the workspace. If it does not, compare the run command, the secrets and the logs before changing any code.